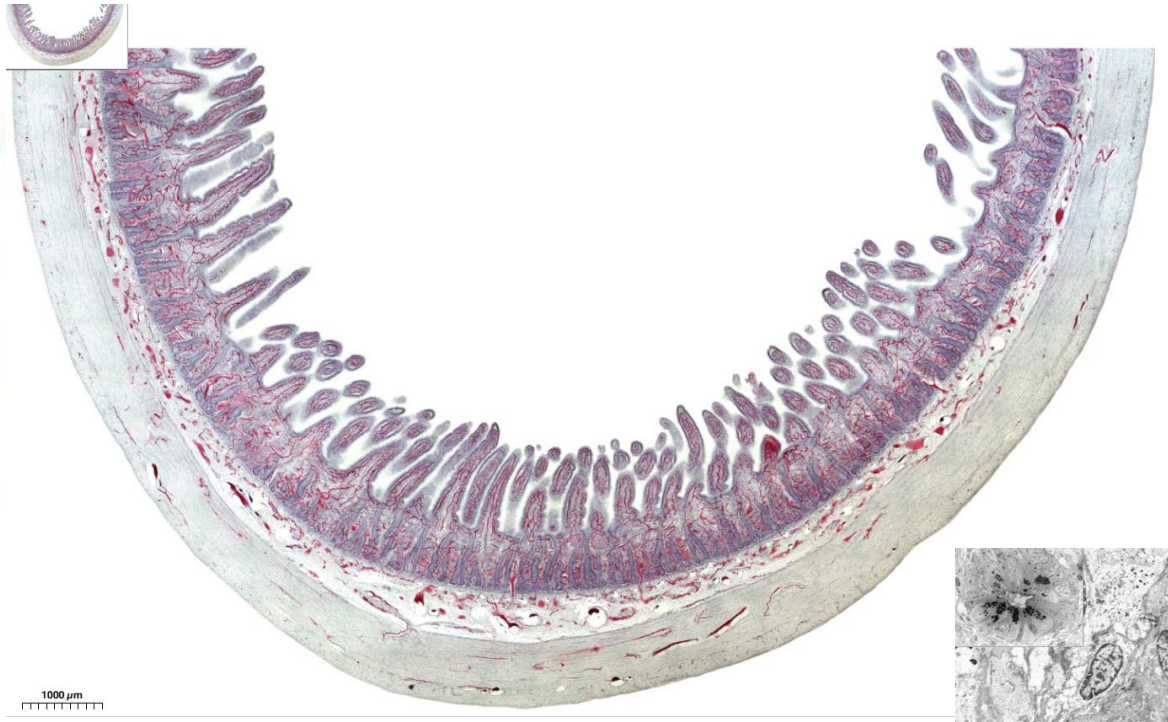
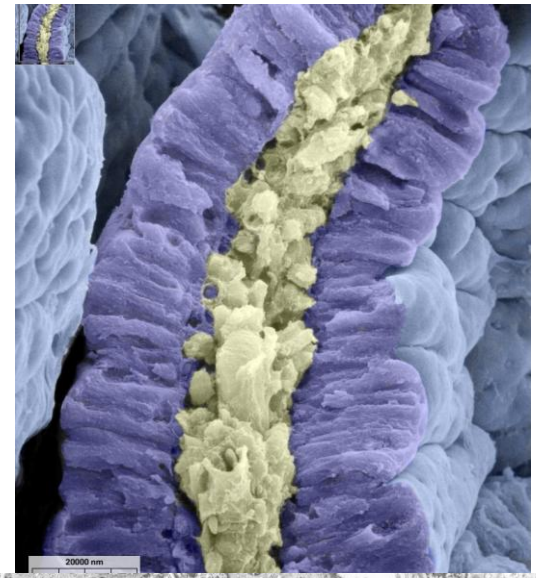
Clinical correlation of the Lower GI Histology







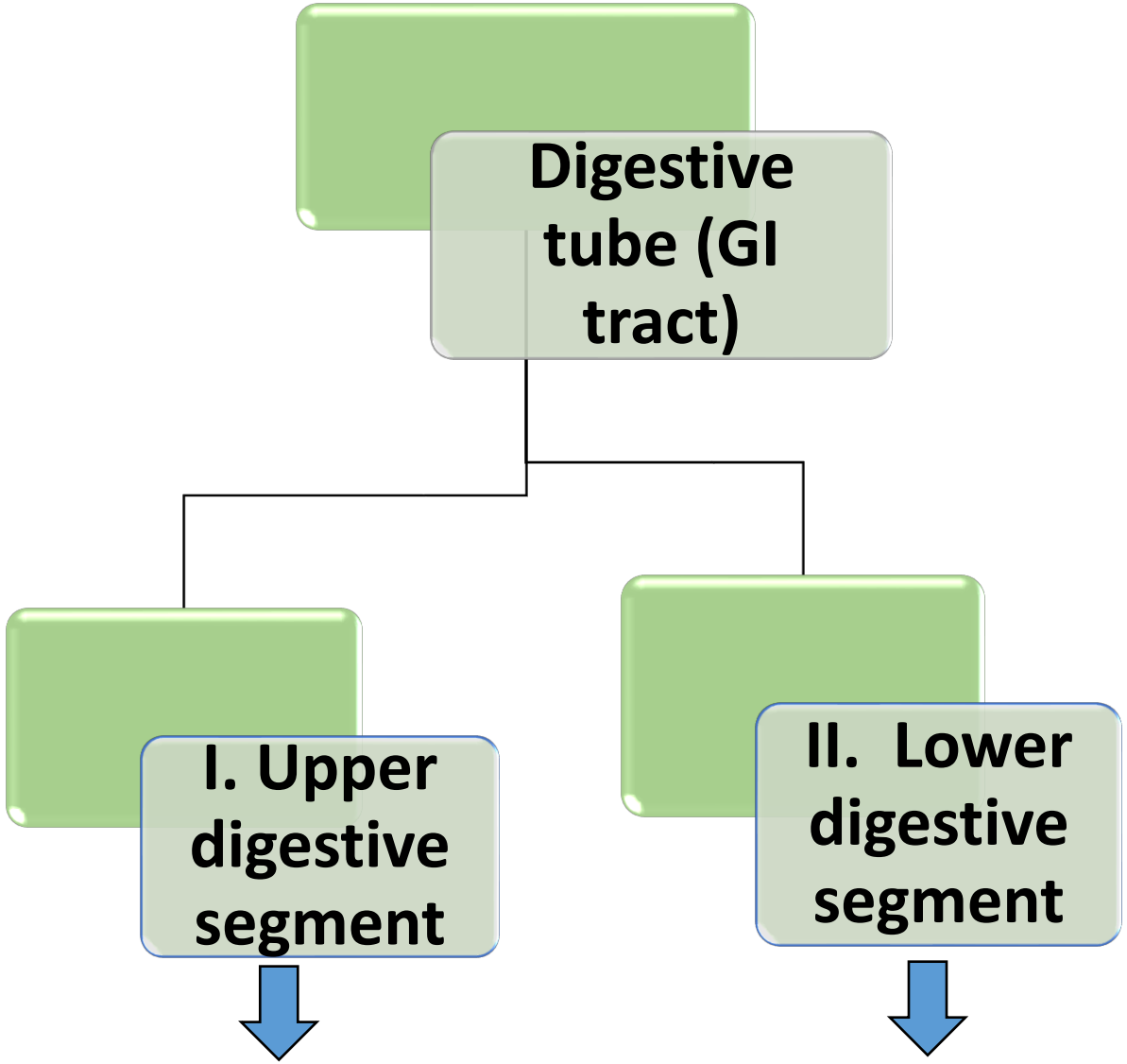






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- **FACULTY OF MEDICINE**
- **CHULALONKORN UNIVERSITY**



Some solid organs in the Mouth & oral cavity(in briefly), **Esophagus, Stomach**

(1)**Small (**i.e., duodenum, jejunum, Ileum**) and** (2)**Large intestine** (i.e., appendix, caecum, colon &rectum)**&**
(3) **Anus**

The main functions of the small intestine are

(1) to continue in the duodenum for the digestive process initiated in the stomach and
(2) to absorb digested food after enzymes produced in the mucosa and pancreas, together with the emulsified bile, enable uptake of protein, CHO and lipid components
(3) To prevent the aggression of pathogens via bacteria, a component of the microbiota and maintain a normal functional relationship with GALT(gut-associated lymphoid tissue)

the large intestine functions mainly to

(1) absorb of water, sodium, vitamins and minerals
(2) secrete mucin (by goblet cells)

↳ Lubricate the feces

Some Clinical Questions to be Addressed for the Lower GI-tract (7 items)

1. Name clinical characteristics of malabsorption syndromes?
2. What is a typical clinical feature in patients of the malabsorption syndromes?
3. What are the causes of the malabsorption syndrome?
4. Name at least 2 inflammatory bowel diseases that account for the defect in the protective system of the lower GI tube. What are the clinical features of 2 inflammatory bowel diseases?
5. What is the Celiac Sprue?
6. What is the clinical features of Hirschsprung's disease? How does to confirm the diagnosis of Hirschsprung's disease in a patient?
7. What is likely to be the precursor structure of the colorectal tumors?

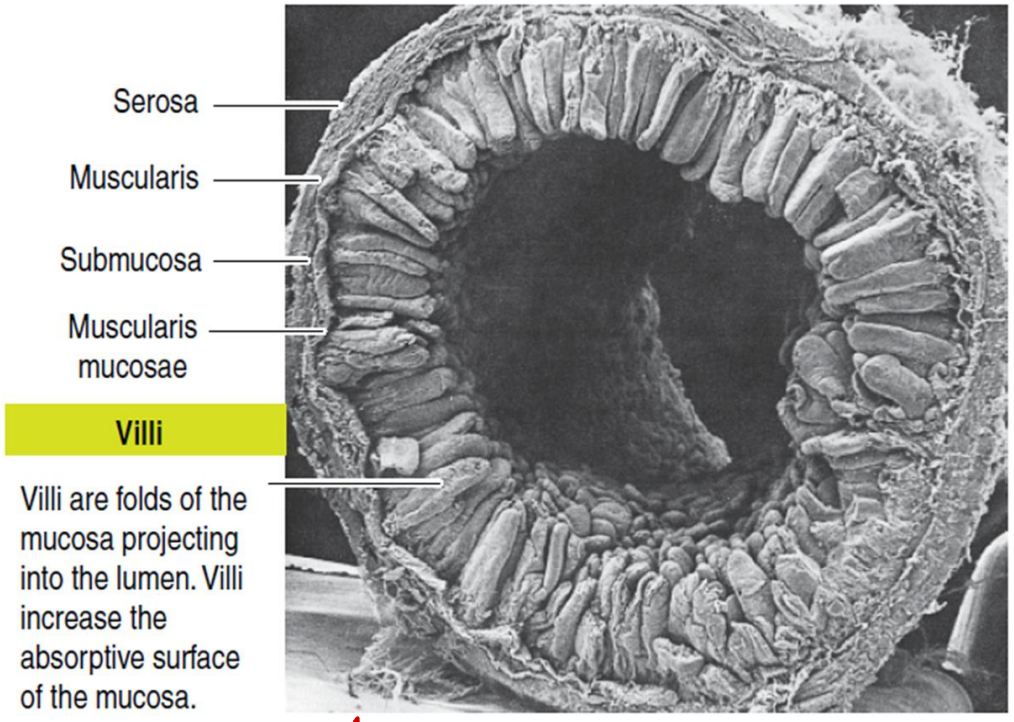
I. SMALL INTESTINE: The 4- to 7-meter-long small intestine is divided into three consecutive segments:

1. **Duodenum**.
2. **Jejunum**.
3. **Ileum**.

The duodenum is about 25 cm in length, is mainly retroperitoneal and surrounds the head of the pancreas. At its distal end, the duodenum is continuous with the jejunum, a movable intestinal segment suspended by a mesentery. The ileum is the continuation of the jejunum. The wall of the small intestine consists of four Layers:

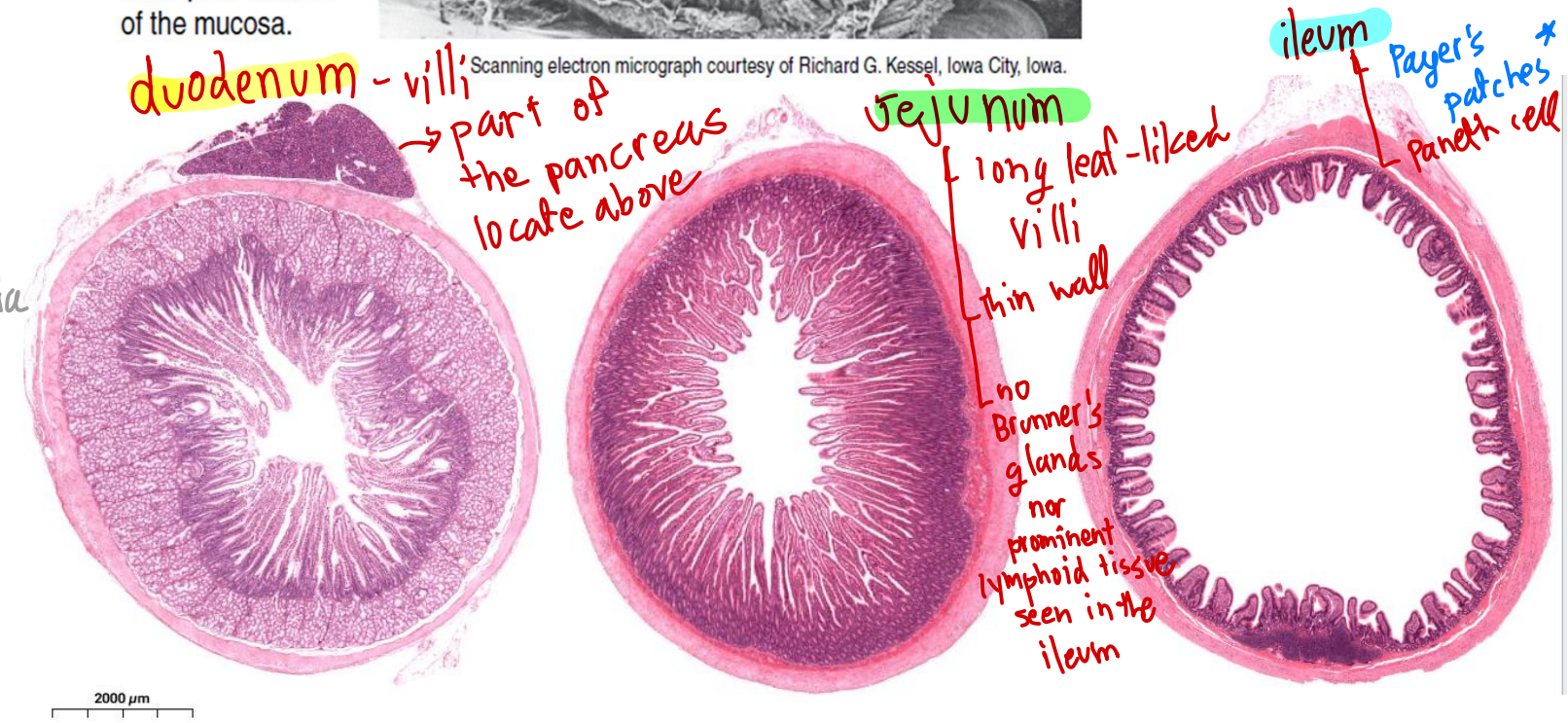
Intestinal gland - in lamina propria
aka crypts of Lieberkuhn

1. The **mucosa**.
2. The **submucosa**.
3. The **muscularis**.
4. The **serosa**, or **peritoneum** / adventitia

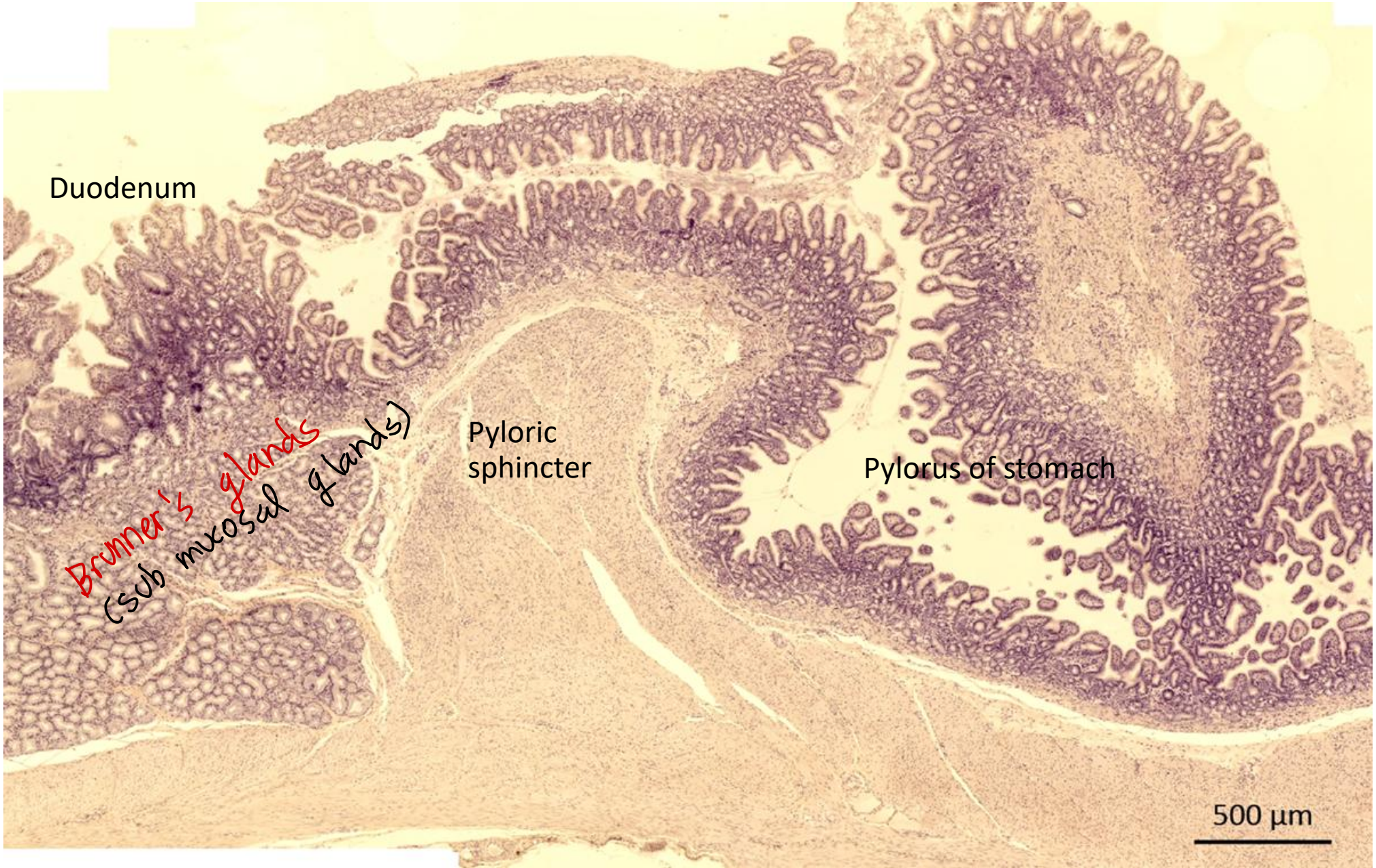


Scanning electron micrograph courtesy of Richard G. Kessel, Iowa City, Iowa.

The intestinal wall shows an increase in the total surface of the *mucosa that reflects the absorptive function* of the small intestine



GASTRO-DUODENAL JUNCTION



Four degrees of folding amplify the absorptive surface area of the mucosa of the small intestine

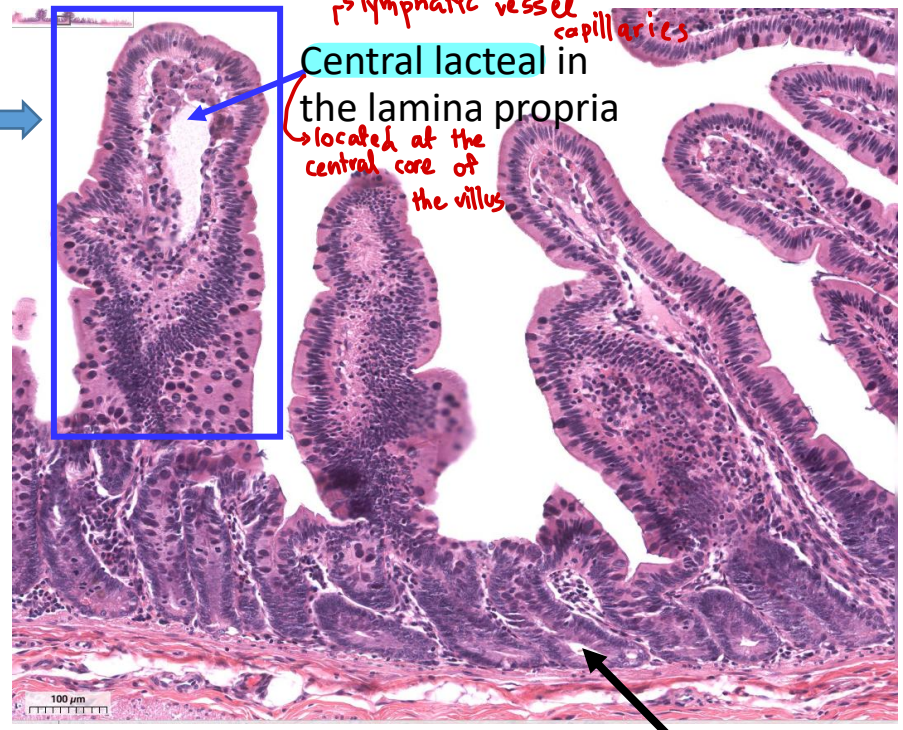
1. *First degree of folding* **plicae circulares** (circular folds; also known as the **valves of Kerkring as submucosal folds**).
2. *Second degree of folding* **intestinal villi**.
3. *Third degree of folding* **intestinal glands** or crypts of lieberkuhn.
4. *Fourth degree of folding* **microvilli** on the apical surface of the lining epithelium of the intestinal cells enterocytes.

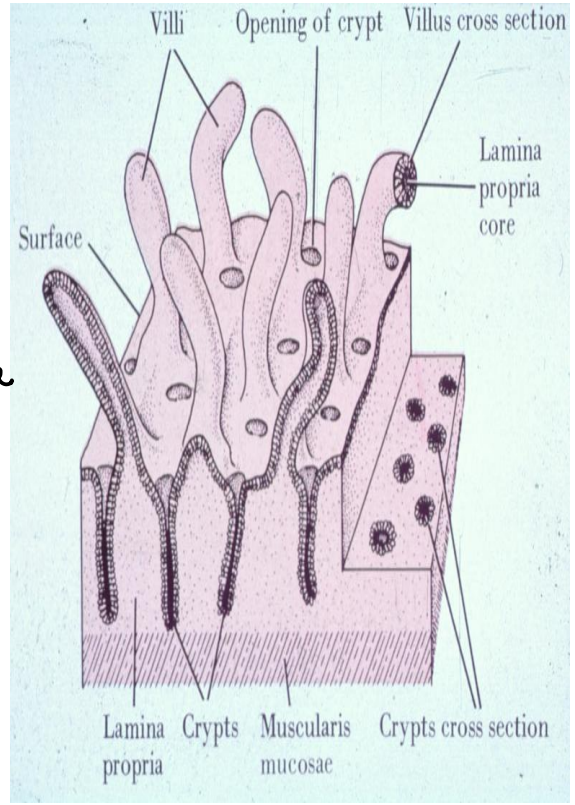
→ absorptive cell



simple columnar epithelium

Villus (lamina propria as a core of villus)







Intestinal gland is surrounded by lamina propris

(in stomach sub mucosal fold " Ruga)

plicae circulares as circular-sub mucosal fold

6 Cell-types in INTESTINAL GLANDS of the SMALL INTESTINE (crypts of Lieberkühn)

1. COLUMNAR ABSORPTIVE CELLS (Enterocytes) with the brush border, a surface coat as glycocalyx (↳ contain enzyme involve in digestion) covering each microvilli, secrete antimicrobial proteins(AMPs) retaining in the intestinal mucus blanket produced by goblet cells
2. GOBLET CELLS, mucous-secreting cells forming a double layer protective gel coat to shield the epithelium as intestinal mucus blanket

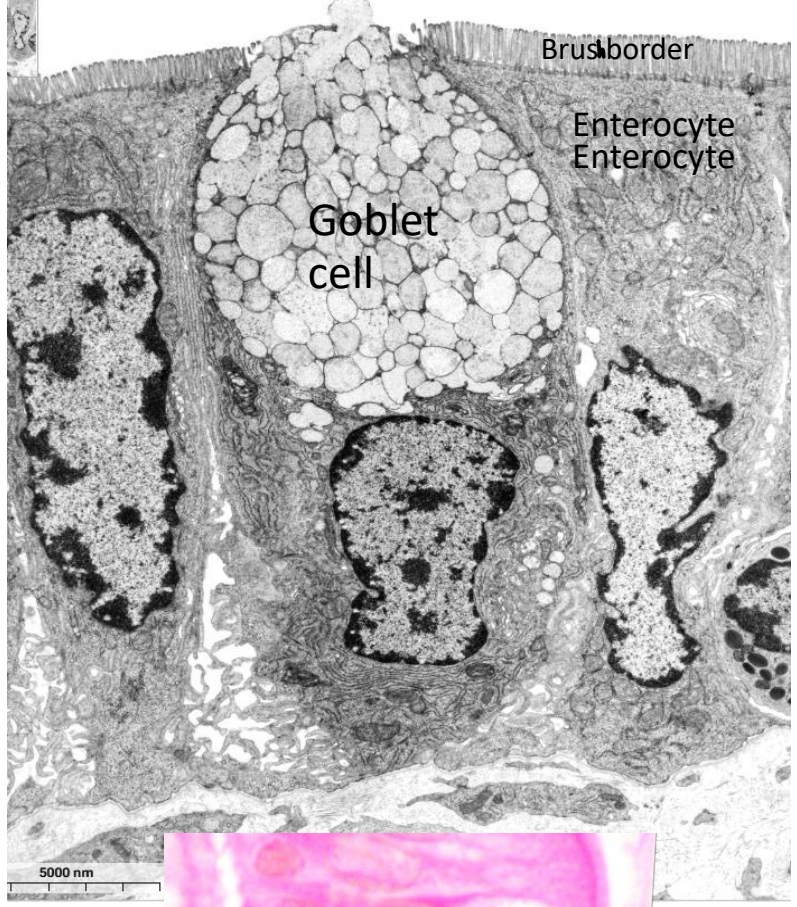
3. PANETH CELLS, at the base of glands in Jejunum & mostly in ileum: secrete antimicrobial proteins(AMPs) retaining in the intestinal mucus blanket produced by goblet cells

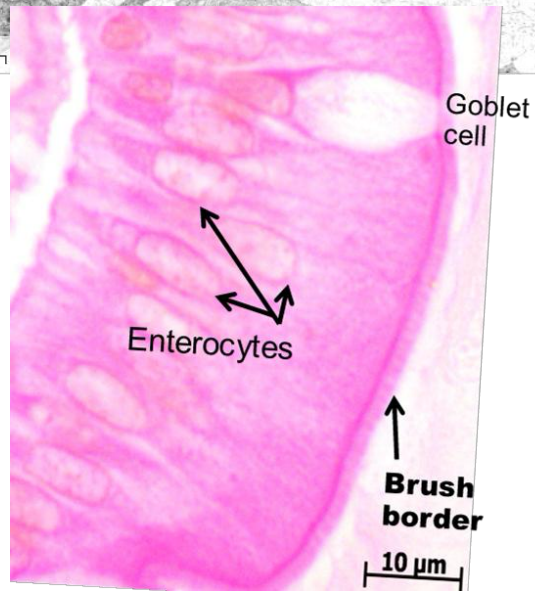
4. ENTEROENDOCRINE (APUD) CELLS (hardly seen in H & E section): secrete peptides hormones

5. TUFT CELLS: a small population of intestinal epithelial cells (o.4%), are flask shaped with a microvillus tuft extending into the intestinal lumen. They increase in number in response to intestinal parasite infection

6. INTESTINAL STEM CELLS (ISC, not seen)) reside in a nich at the base of the crypts, close to Paneth cells

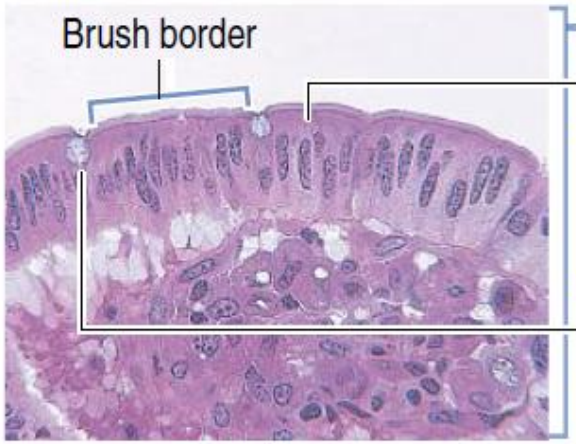
L = intestinal gland

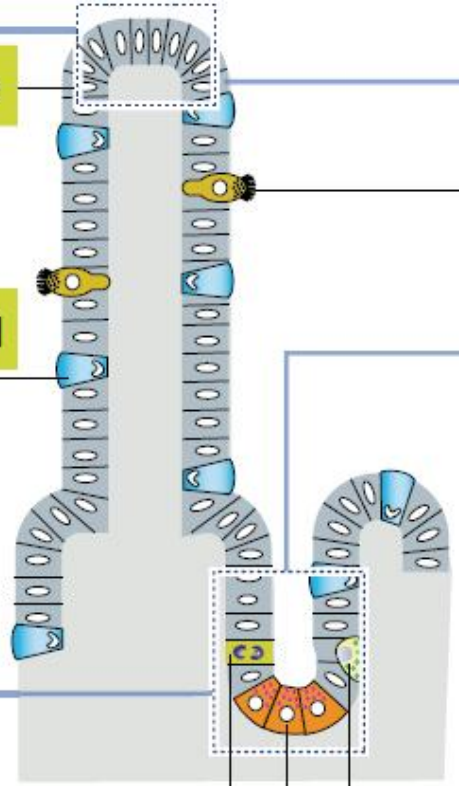


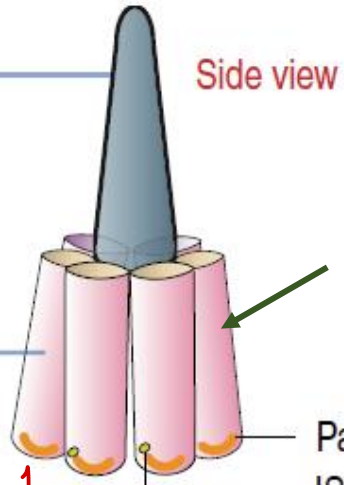


Epithelial cells covered villus (has lamina propria as a core) & about 6 crypts of Lieberkuhn or intestinal glands(are surrounded by lamina propria)

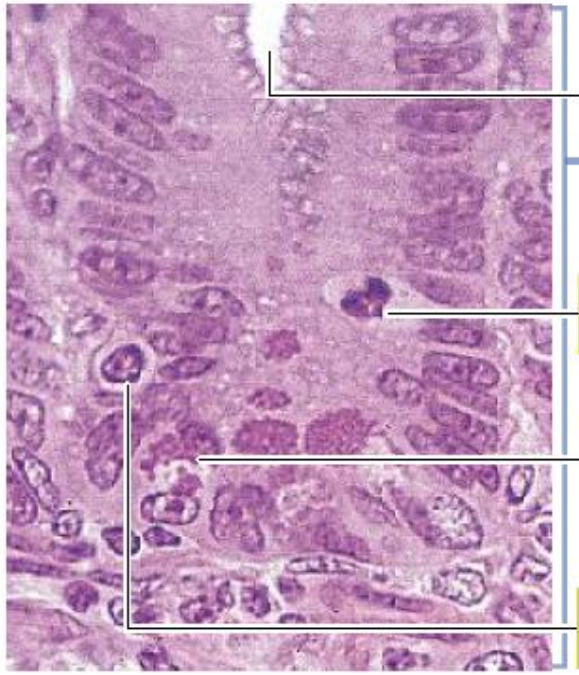
↳ surround single villus





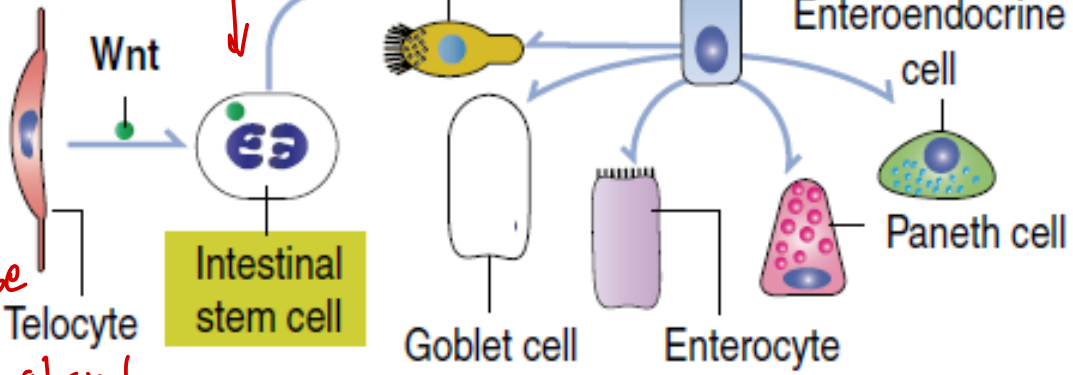


each intestinal villus is surrounded by at least six crypts of Lieberkuhn.



located at the base of the intestinal gland

all cells derived from here



A subset of mesenchymal-derived dendritic cells in the lamina propria, called **telocytes**, is juxtaposed to the crypts or glands of Lieberkuhn.

Histological features of cell-types in the mucosal small intestine

digest food into absorbable nutrients

I. Absorptive cells or enterocytes



TEM OF

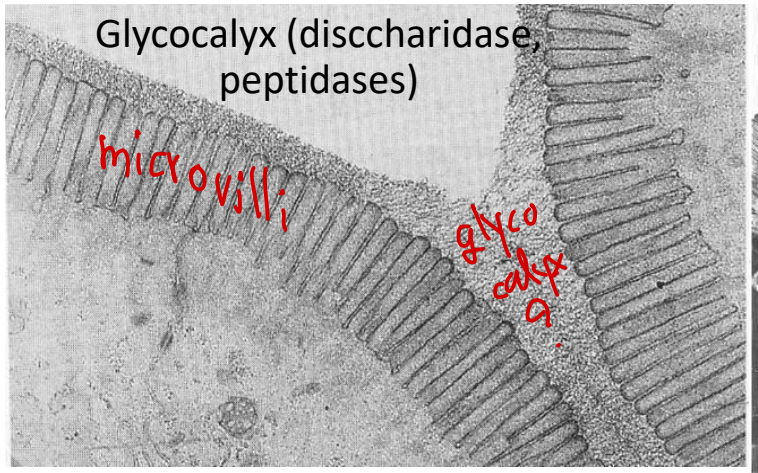
AN ABSORPTIVE CELL *or* ***ENTEROCYTES****, have*

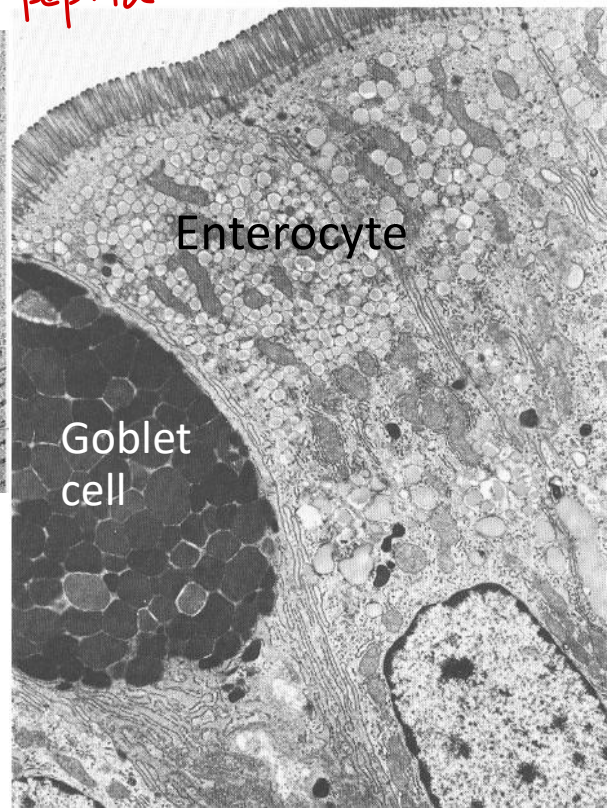
(1) Microvilli :increasing 30 folds surface

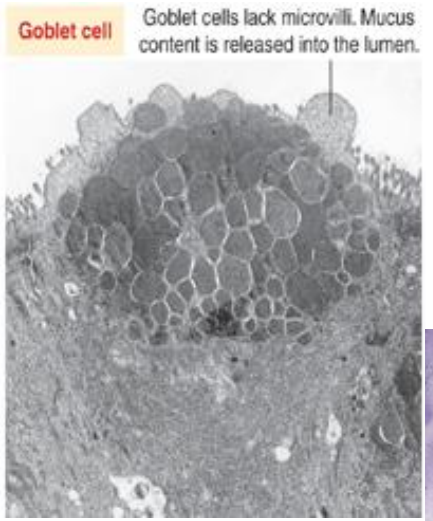
(2) Anchorage of enzyme (in the glycocalyx, involved in digestive process) & intrinsic factor receptors

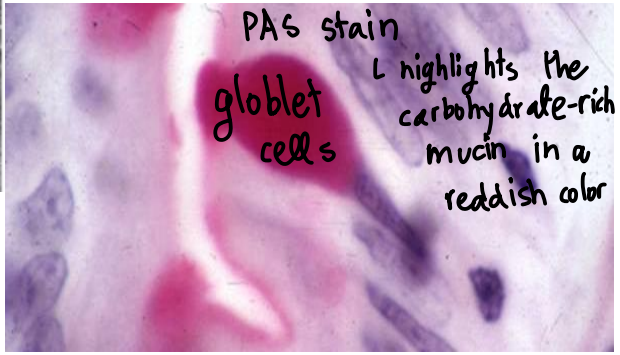
(3) Produce AMP

attachment site of various enzyme ex. saccharidases & peptidases

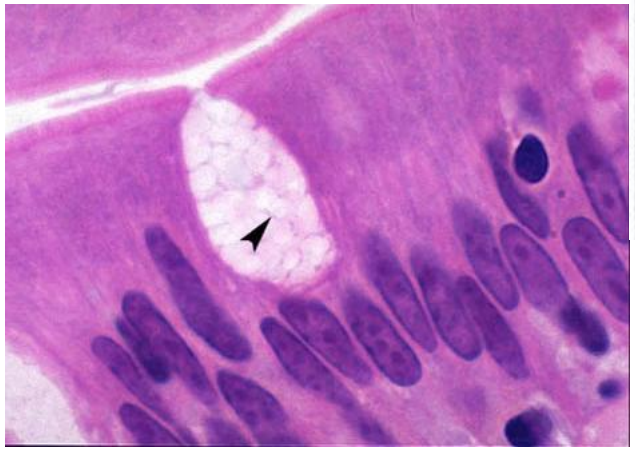








*II. **GOBLET CELLS** SECRETE MUCIN, GLYCOPROTEINS assembled into a vicious gel-like blanket limiting direct bacteria contact with enterocytes.*



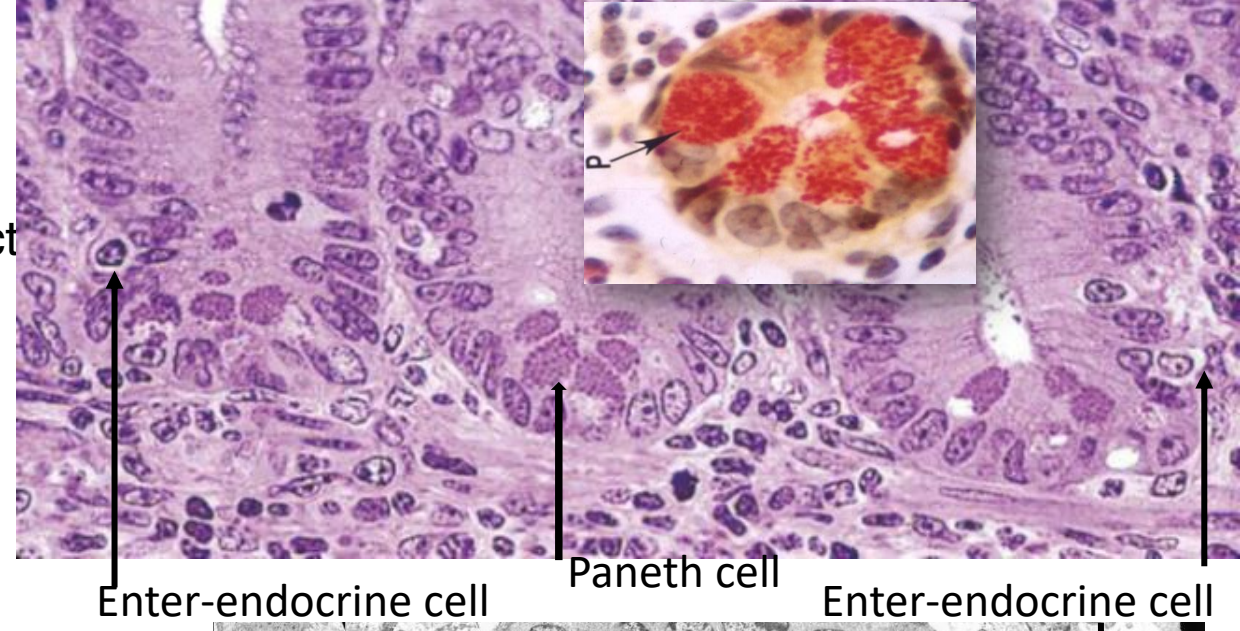
III. Entero-endocrine cells, i.e.,

1. **Secretin cells** (in duodenum):

1) It stimulates **bicarbonate** secretion by the pancreatic duct
2) It enhances **insulin** secretion by B cells of the islet of Langerhans

2. **Cholecystokinin cells** (in duodenum)

1) It stimulates **bile release** from the gallbladder and the secretion of **pancreatic enzymes**.
2) It slows down emptying of the stomach by acting on the **pyloric sphincter**



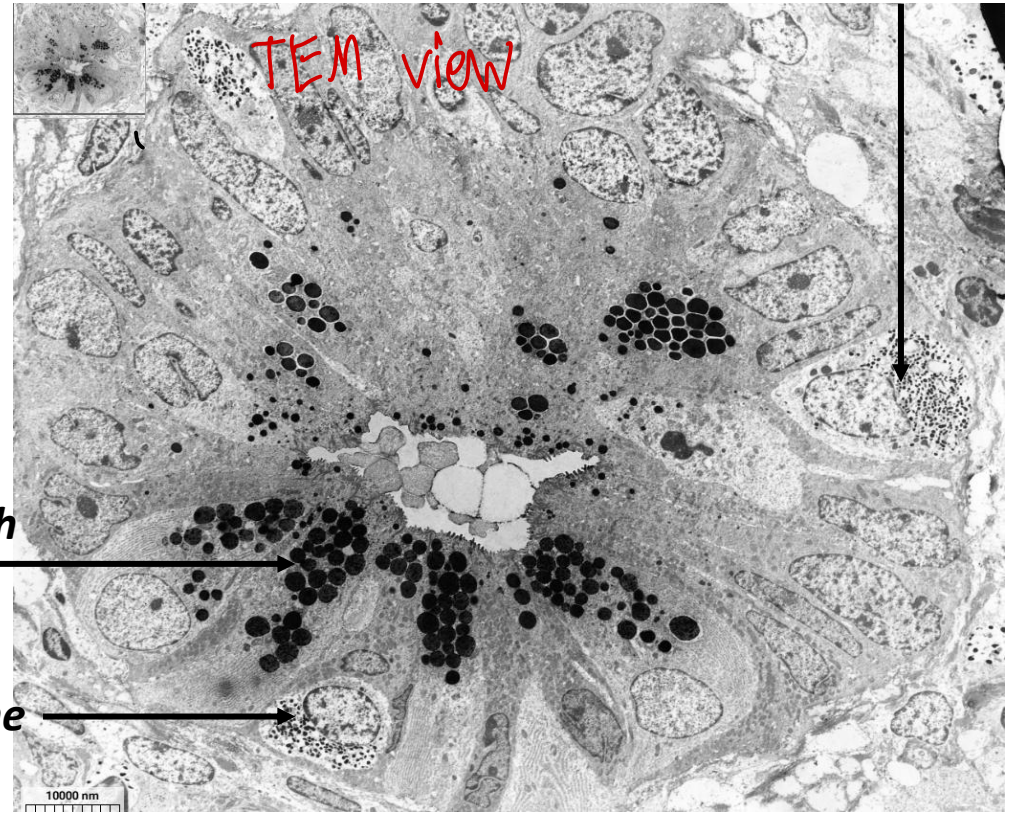
→ produce and release granule directly through the apical surface

IV. Paneth cells are present at the base of the crypts of Lieberkuhn (in close proximity to the ISCs and have a life time of about 3 to 6 weeks, in contrast with the rapid turnover of the enterocytes).

The pyramid-shaped Paneth cells have a basal domain containing the RER. The apical regions show numerous protein granules representing a diverse array of **antimicrobial proteins** (**AMPs**), an indication of the microbial diversity and impending threats.

Enterocytes also produce AMPs.

destroying bac enter intestine



In the case of these endocrine cells viewed with TEM

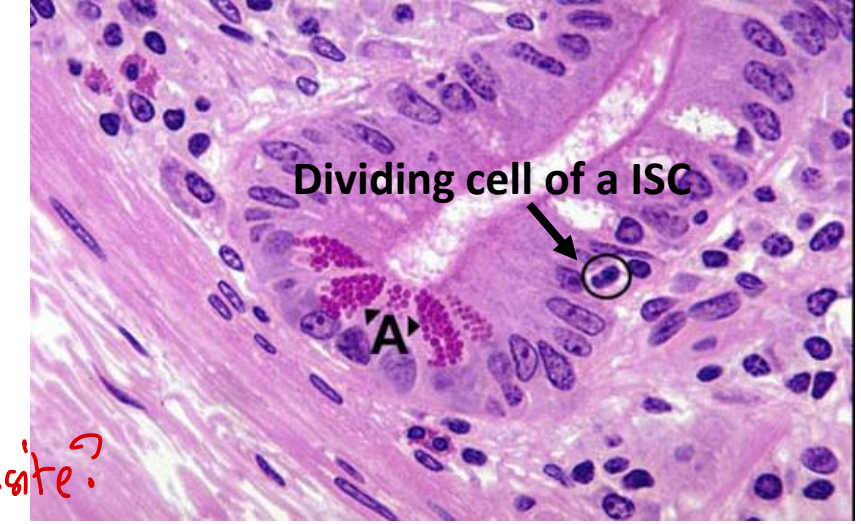
- The granules are smaller and are located in the basal cytoplasm, below the nucleus because hormones are not released through apical surface, They diffuse out from the basal cell membrane and puss through the microcirculation within the lamina propria

V. Intestinal stem cell(ISC)niche in the crypt of Lieberkuhn, located at the base of the intestinal crypt, give rise to:

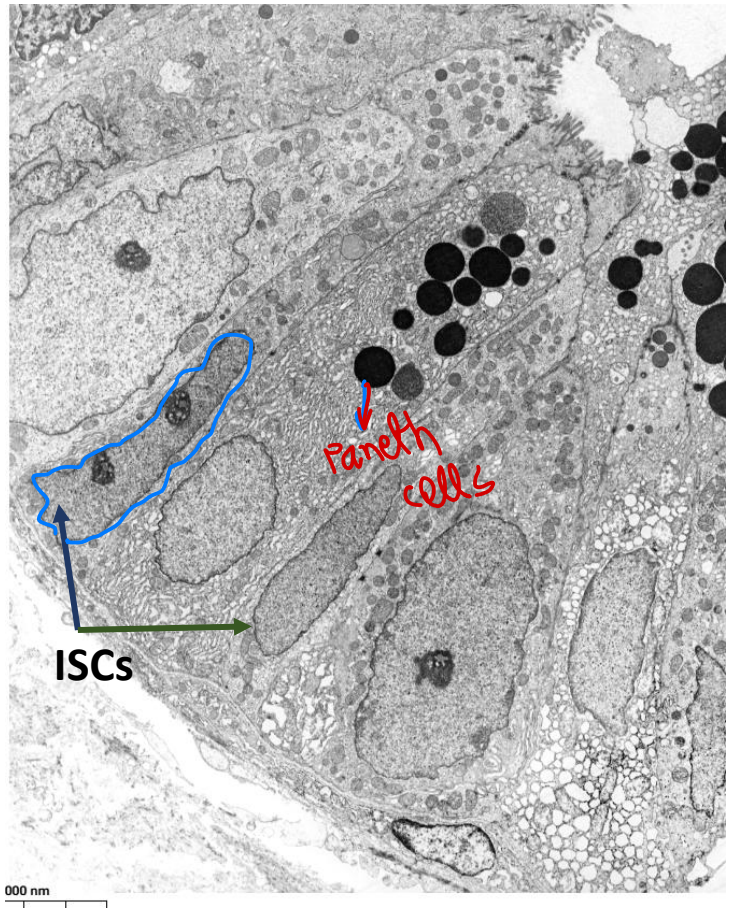
(1) Paneth cells, which remain close to ISCs at the base of the intestinal crypt.

(2) Enterocytes, goblet cells and enteroendocrine cells,

which move out of the crypt to replace the epithelial lining of the villus. Epithelial cell turnover occurs every 3-5 days. New Paneth cells are produced every 3-6 weeks



against parasite?

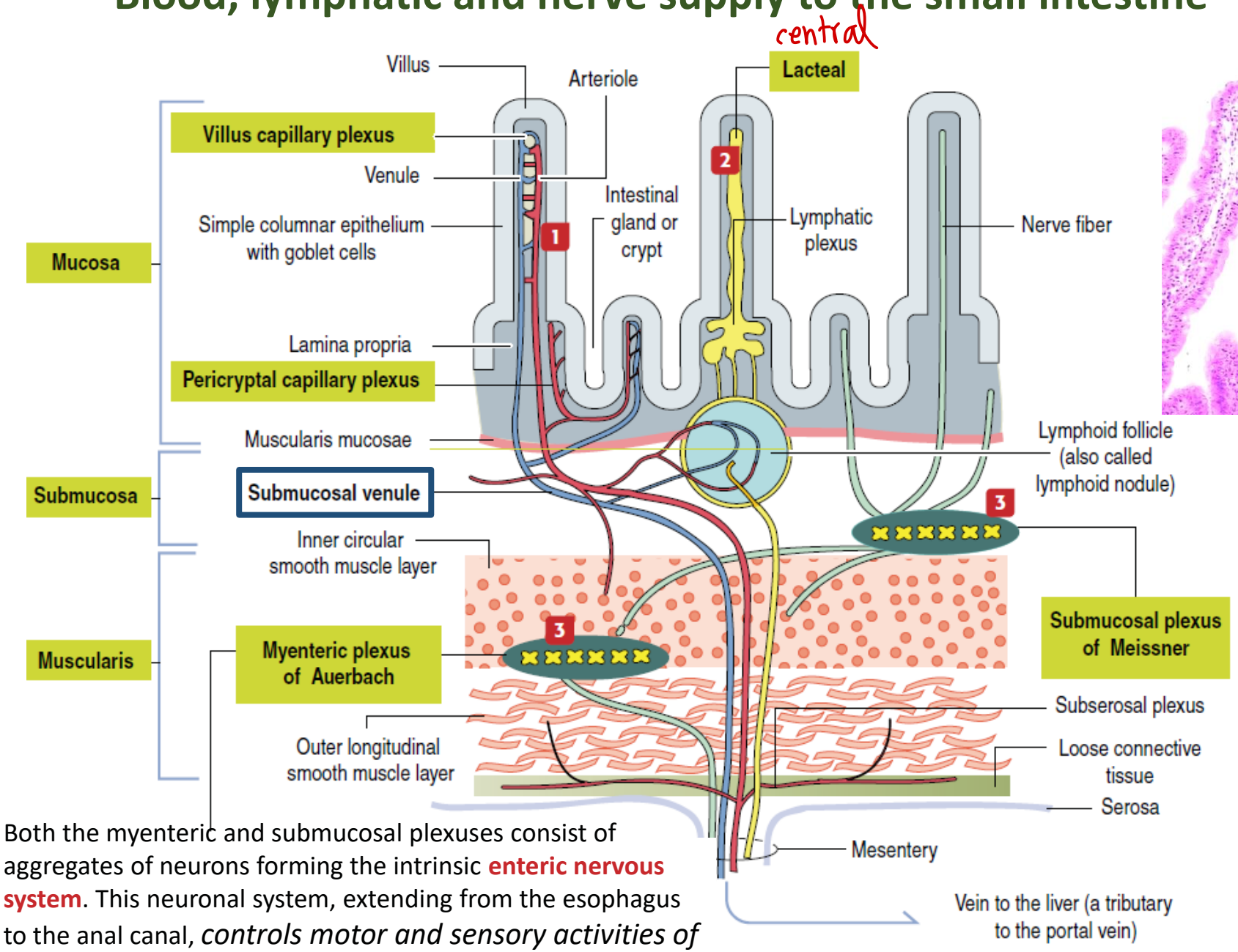


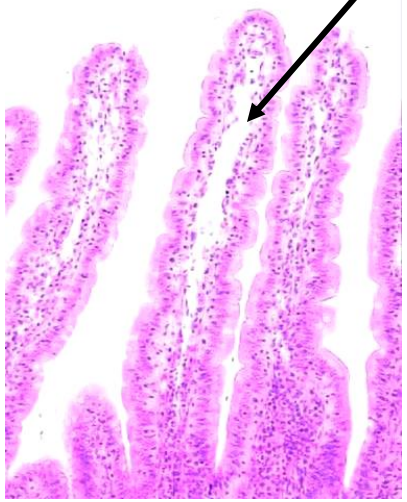
* *VI. Tuft cells* as a small population of intestinal epithelial cells (about 0.4%), are flask shaped with a microvillus tuft extending into the intestinal lumen. Clusters of vesicles are seen at the base of the microvillus tuft.

Tuft cells are responsible for initiating responses to parasitic infection by increasing in number, stimulating an expansion of the population of goblet cells and stimulating the production of **interleukin-25** by an unidentified intestinal epithelial cell. Following a 7-day post-infection period, the anti-helminth immune response, initiated by tuft cells, results in parasite expulsion.

* *No both LM & TEM shown in the histological textbooks, only the drawing tuft cells*

Blood, lymphatic and nerve supply to the small intestine





A single blind-ending **central lymphatic capillary**, called a **lacteal**, is present in the core or lamina propria of a villus

The microvascular system of the villus derives from two arteriolar systems. One system supplies

(1)the tip of the villus as **villus capillary plexus**
(2). the **peri-cryptal capillary plexus**.

Both plexuses drain into the **submucosal venule**

Both the myenteric and submucosal plexuses consist of aggregates of neurons forming the intrinsic **enteric nervous system**. This neuronal system, extending from the esophagus to the anal canal, *controls motor and sensory activities of the digestive tube.*

Histological features of

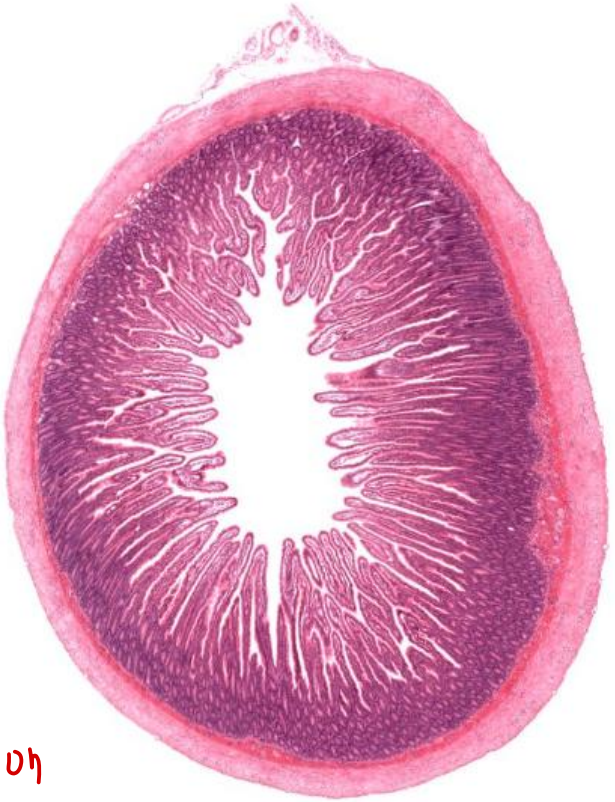
1. DUODENUM



- Pancreatic duct also open into this region

1. Villi with a leaflike shape
2. Mucus-secreting Brunner's glands in the submucosal glands as tubulo-acinar mucous gland produce an alkaline secretion that neutralizes the acidic chyme
3. Excretory ducts of Brunner drain into the crypts of Lieberkuhn

2. JEJUNUM

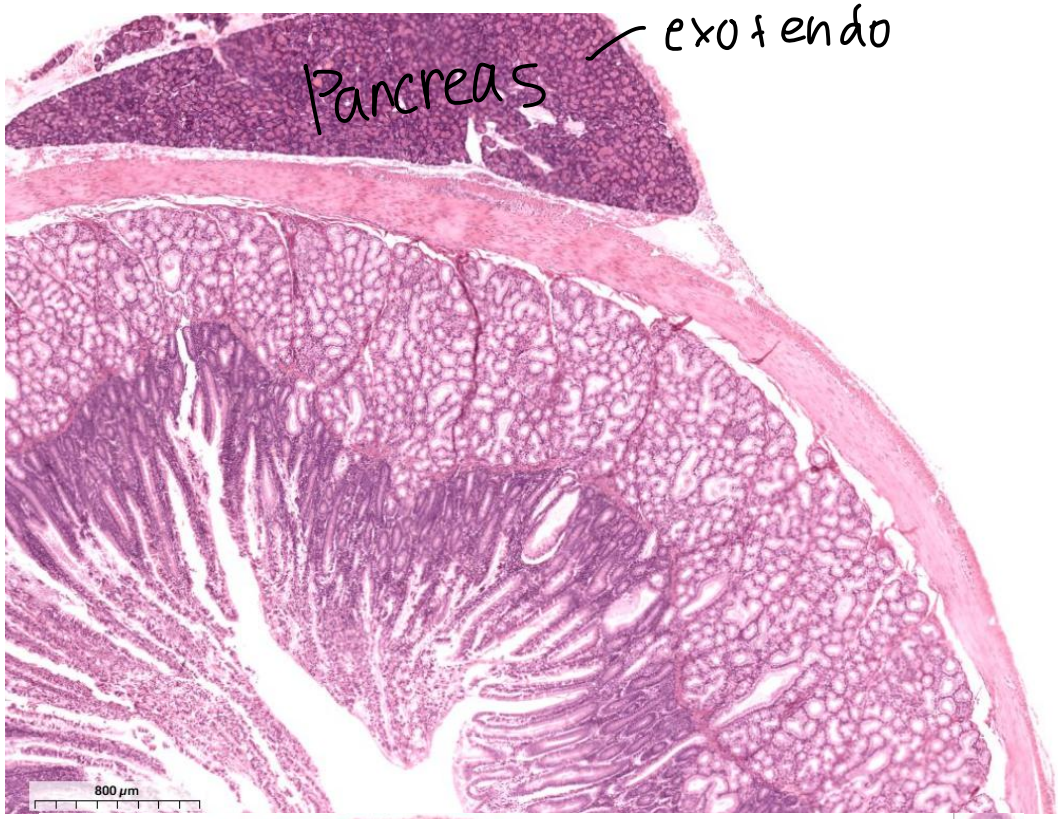


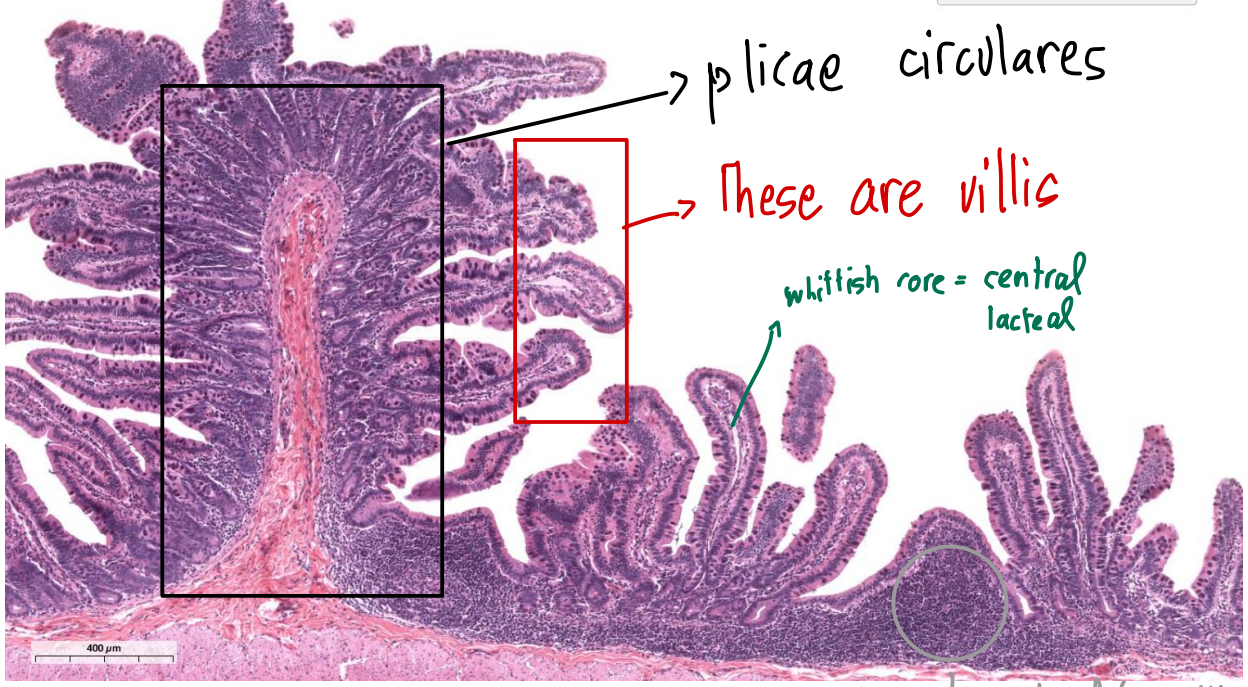
1. Villi have a finger-like shape
2. Plicae circulares are well developed
3. No glands in the submucosa

3. ILEUM

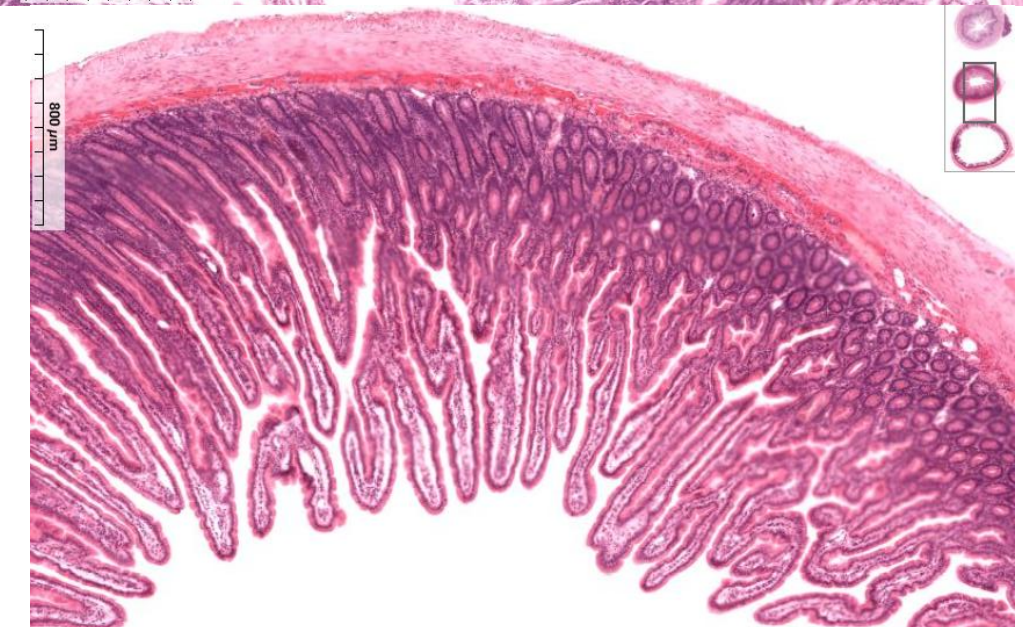


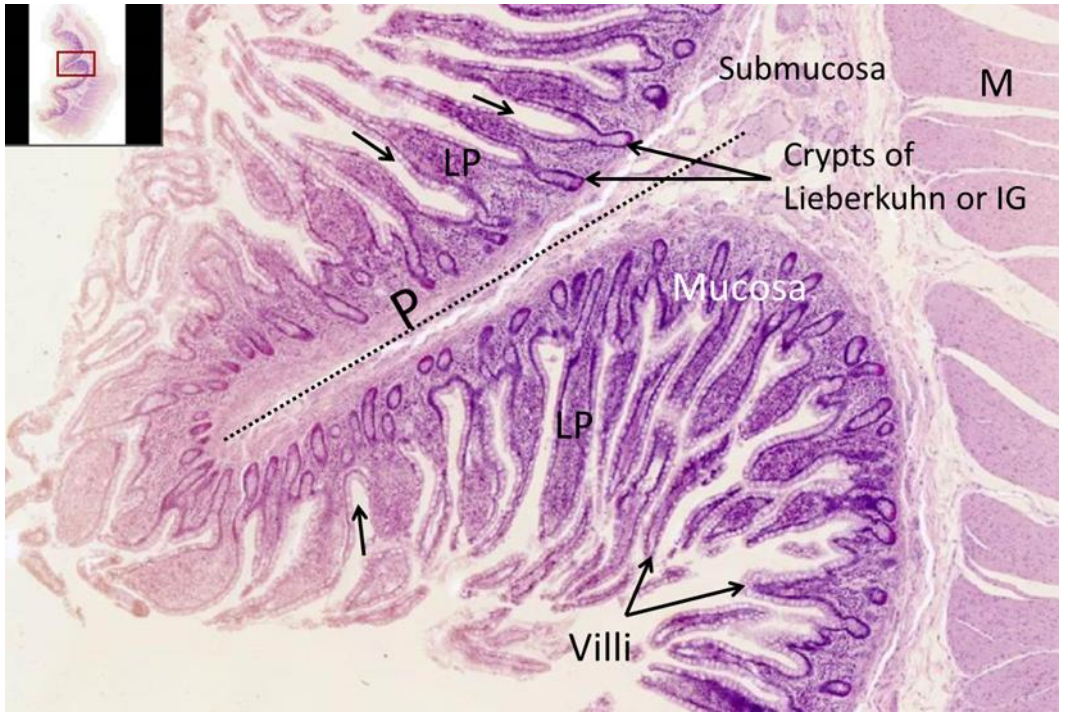
1. Villi are shorter when compared with the jejunum → microfold cells
2. M cells and enterocytes are located in the follicle-associated epithelium
3. Lymphoid aggregated known as Peyer's patches(thru lamina propria & submucosa)





Peyer's patches with epithelium covering over them





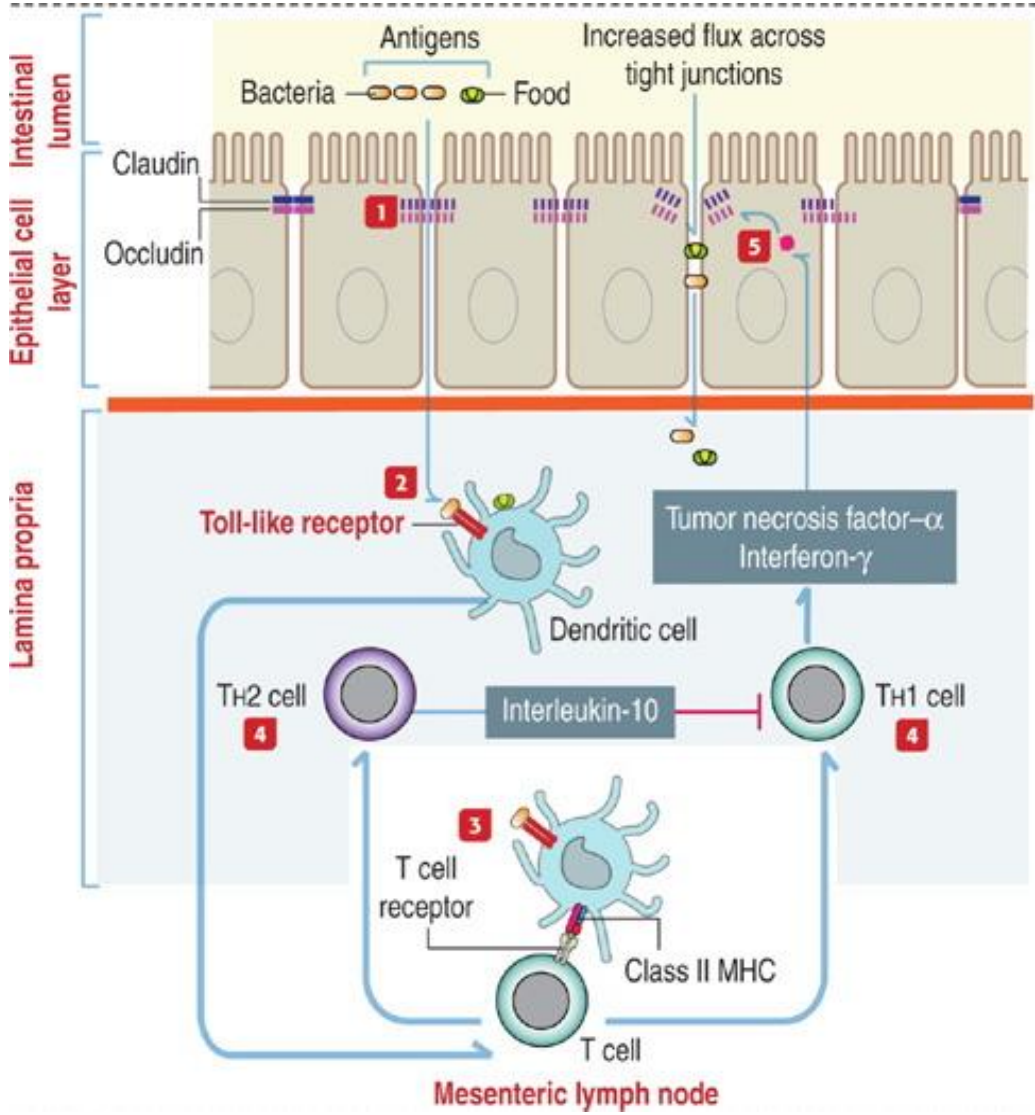
In the small and large intestines, **goblet cells** secrete mucin glycoproteins assembled into a viscous gel-like blanket, limiting direct bacterial contact with enterocytes

The small intestine is protected from the microbiota.
What is the intestinal microbiota? It is a collection of bacteria, archaea, viruses, fungi and parasites, which reside in the intestinal lumen and on the surface of the intestinal mucosa.
The intestinal epithelium forms a defensive barrier between a host and its microbiota.

The components of the defensive barrier include:

(1) **An intestinal tight junction barrier,** linking adjacent enterocytes. Claudin and occludin are two transmembrane proteins of tight junctions, which regulate solute permeability of the transcellular pathway.

- A defect of the tight junction barrier can allow ***bacterial products or dietary antigens*** to cross the epithelium and enter the lamina propria.
- Unregulated inflammatory cytokines signal enterocytes to increase the passage of antigens across leaky tight junctions, from the lumen to the lamina propria, thus amplifying the inflammatory reaction.
  This mechanism can lead to *an intestinal inflammatory disease*

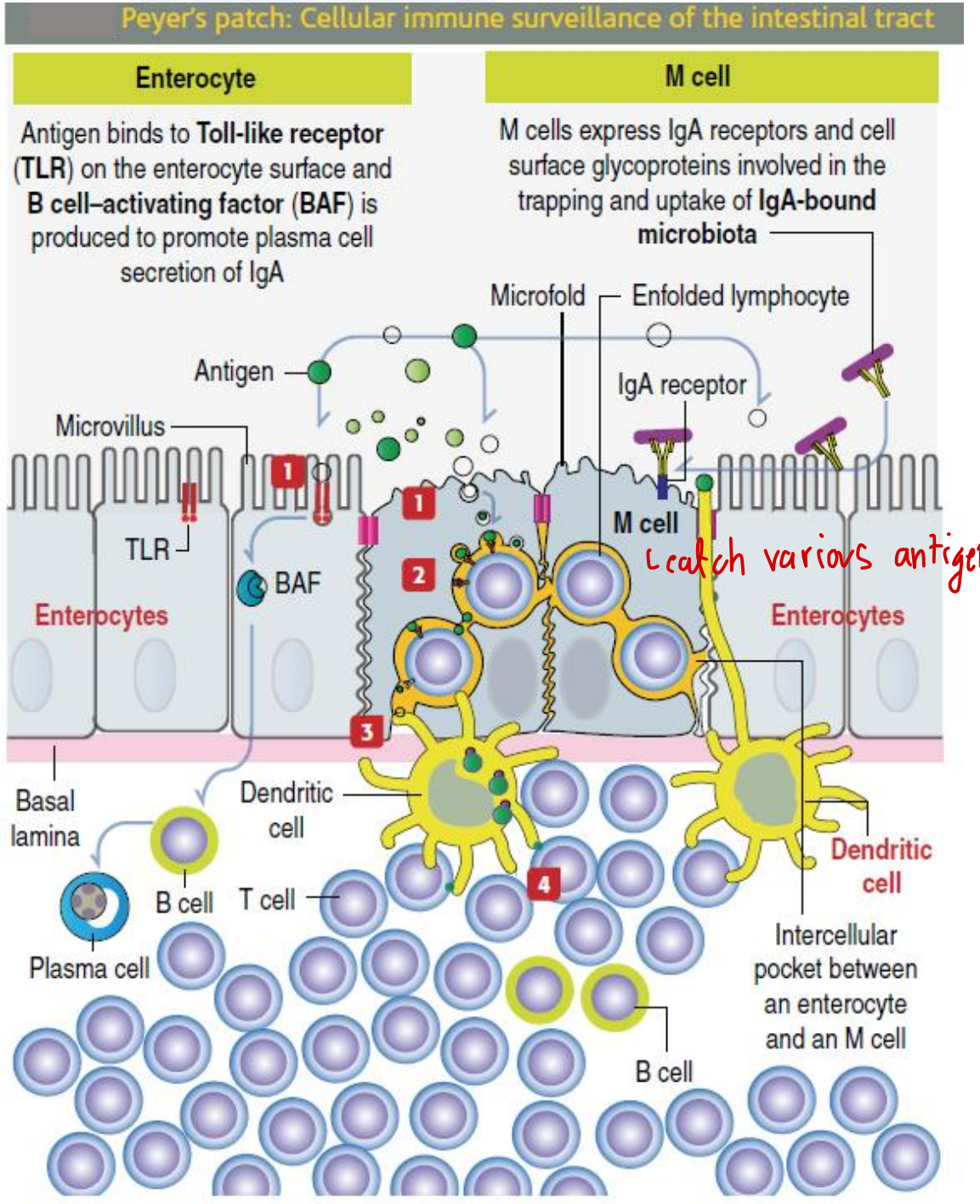


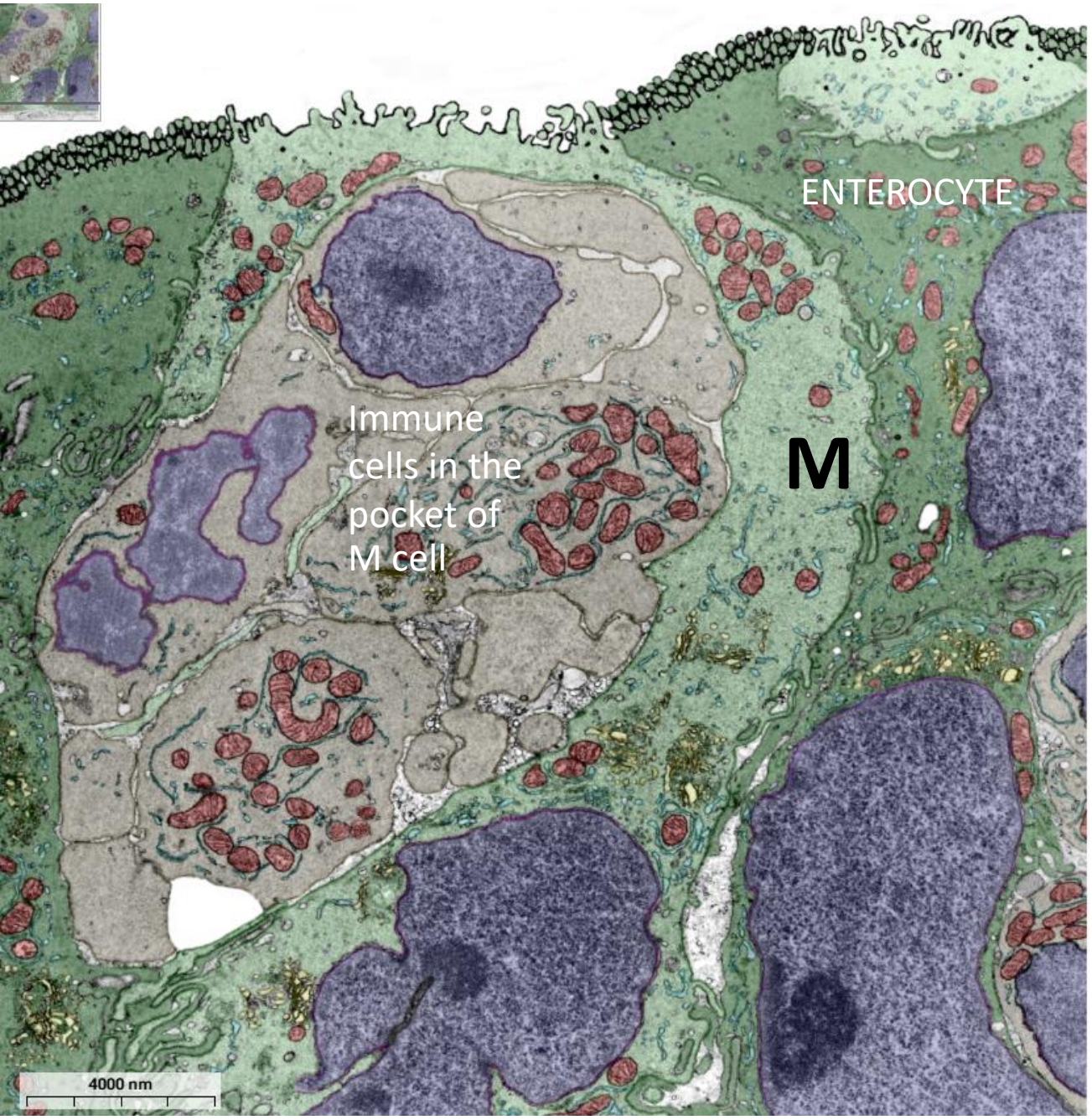
(2) **Peyer's patches** participate in the *cellular immune surveillance and processing of antigens*. Peyer's patches have the ability to transport luminal antigens and microorganisms and respond to them by inducing immune tolerance or a systemic immune defense response.
*An example of the functional deficiency of Peyer's patches is **Crohn's disease**, an inflammatory bowel disease characterized* by chronic or relapsing inflammation.

Peyer's patches consist of three main components:

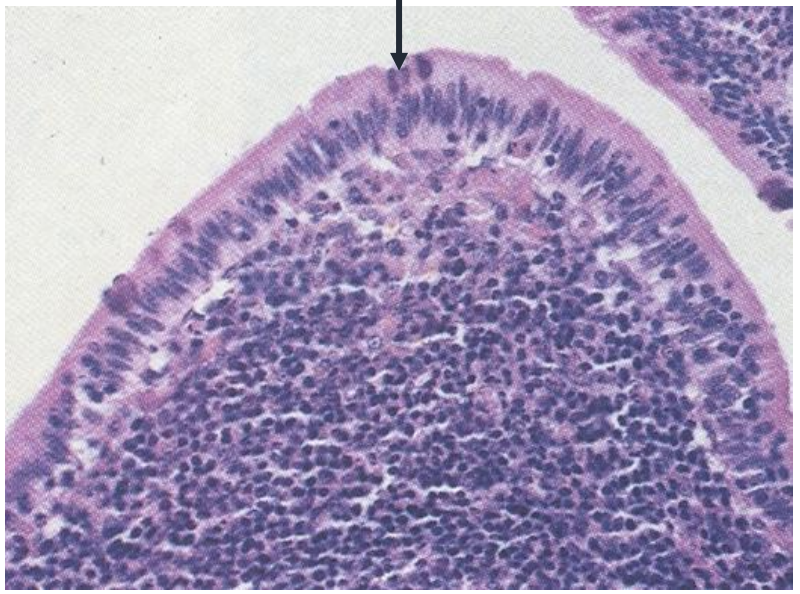
(i) The follicle-associated epithelium (FAE), consisting of M cells and enterocytes
(ii) Lymphoid follicles, each showing a germinal center and a subepithelial dome area.
(iii) The interfollicular area, with blood vessels and efferent lymphatic vessels.

The main components of the FAE are M cells (with a pocket containing immune cells) and enterocytes. M cells are specialized enterocytes, which have replaced the brush border by short microfolds.





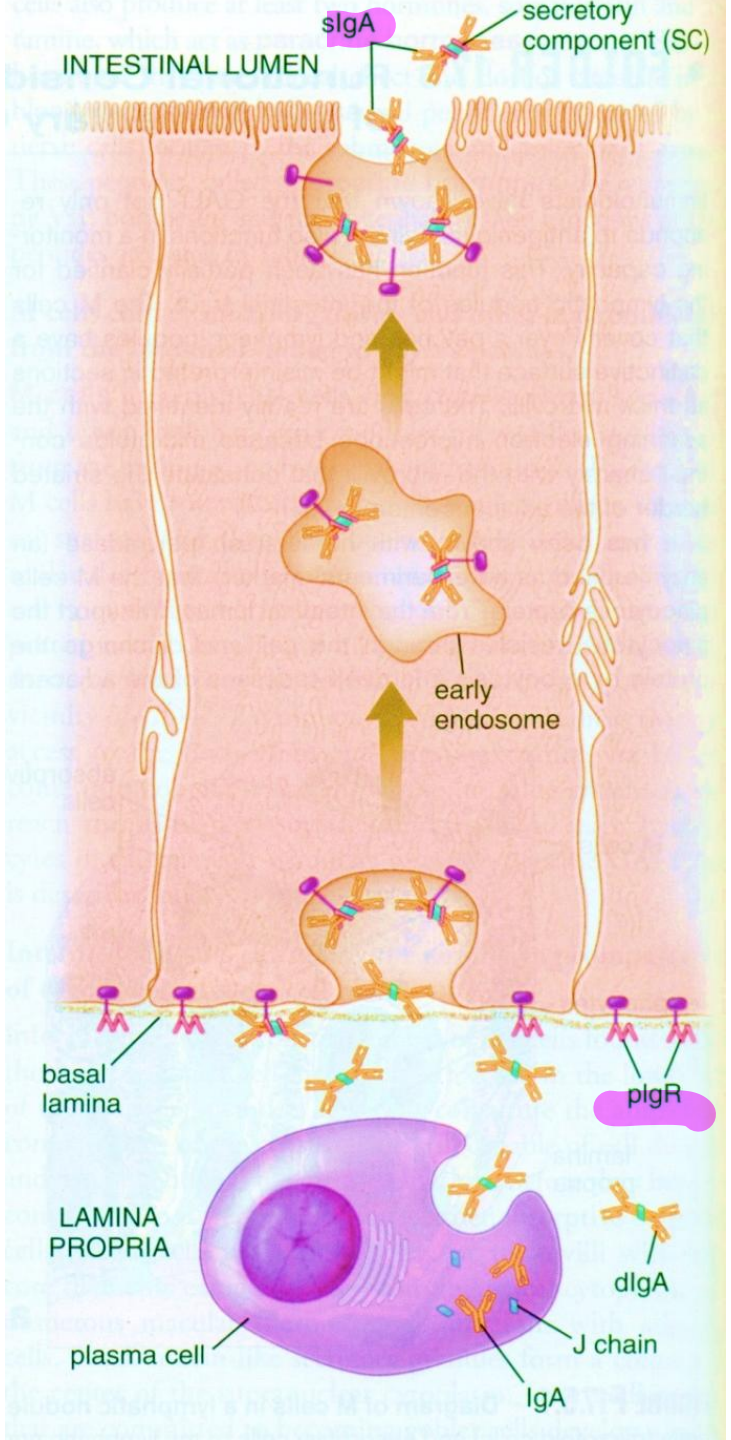
Follicle associated epithelium (FAE)



M (MICROFOLD) CELLS at FAE (Enterocytes &M cells) of Peyer's patch, take up antigens & replaced the brush border by short microfolds

Function of M cells act as antigen-sampling cells. They:

1. Capture bacteria, viruses, and other antigens from the intestinal lumen.
2. Transport these antigens across the epithelial cell by transcytosis.
3. Deliver the antigens to dendritic cells, macrophages, and lymphocytes in Peyer's patches.
4. Initiate mucosal immune responses, leading to the activation of B cells and the production of secretory IgA (SIgA).



(3.) IgA dimers :The molecular immune surveillance of the intestinal tract as followings

1. Plasma cells produce polymeric IgA (pIgA), in which two IgA molecules are joined by a J chain to form dimeric IgA (dIgA).
2. Dimeric IgA (dIgA) binds to the polymeric immunoglobulin receptor (pIgR) on the basolateral surface of enterocytes.
3. The pIgA–pIgR complex undergoes *transcytosis* to the apical surface of the enterocyte, where it is released as secretory IgA (SIgA).
4. Secretory IgA (SIgA), consisting of IgA bound to the secretory component, is released into the intestinal lumen.
5. SIgA binds to bacteria and other antigens in the intestinal lumen, preventing their attachment to the epithelium and facilitating immune surveillance by M cells and other mucosal immune cells.

Ig A attaches to bacteria and soluble antigens, preventing a direct damaging effect to intestinal cells and penetration into the lamina propria.

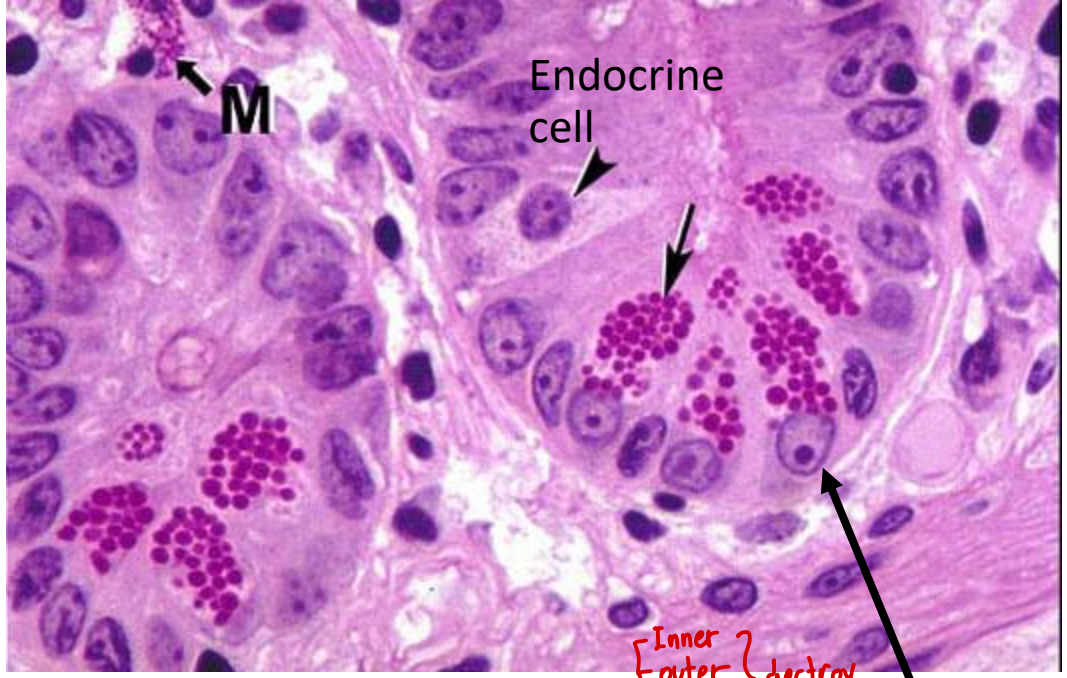
The excess IgA dimers diffuse from LP into bloodstream & are excreted into intestinal lumen via the bile

(4) **Paneth cells** regulate the resident & pathogenic bacterial microflora as adaptive antimicrobial immunity

Intestinal antimicrobial proteins(AMPs)
: Most of AMPs inactivate or kill bacteria directly by enzymatic degradation of bacterial wall or disrupting the bacterial inner membrane.
:AMPs produced by *Paneth cells* and *enterocytes* are retained in the intestinal mucous blanket produce by goblet cells.

(no image)

(5) **Tuff cells**, sensing the microbiota and triggering an immune response to intestinal parasites.



Paneth cells at the base of the crypts of both in jejunum and ileum

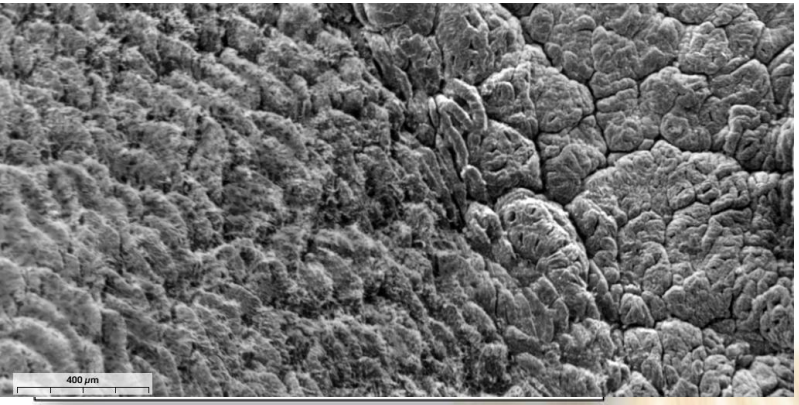
Intestinal mucus blanket (2 layers) Inner / outer } destroy pathogen enter intestine

In the intestine, **goblet cells** secrete a mucus glycoprotein blanket consisting of stratified outer and inner layers. The blanket is deposited on the surface of the intestinal epithelium.
Microorganisms predominate in the outer mucus layer, whereas the inner mucus layer, resistant to microorganism penetration, contains *antimicrobial proteins secreted by Paneth cells and enterocytes*.

mucus layer
① creating a barrier that limits direct access of luminal bac to the epitheli-um
② concentrating antimicro-bial proteins near the enterocyte surface

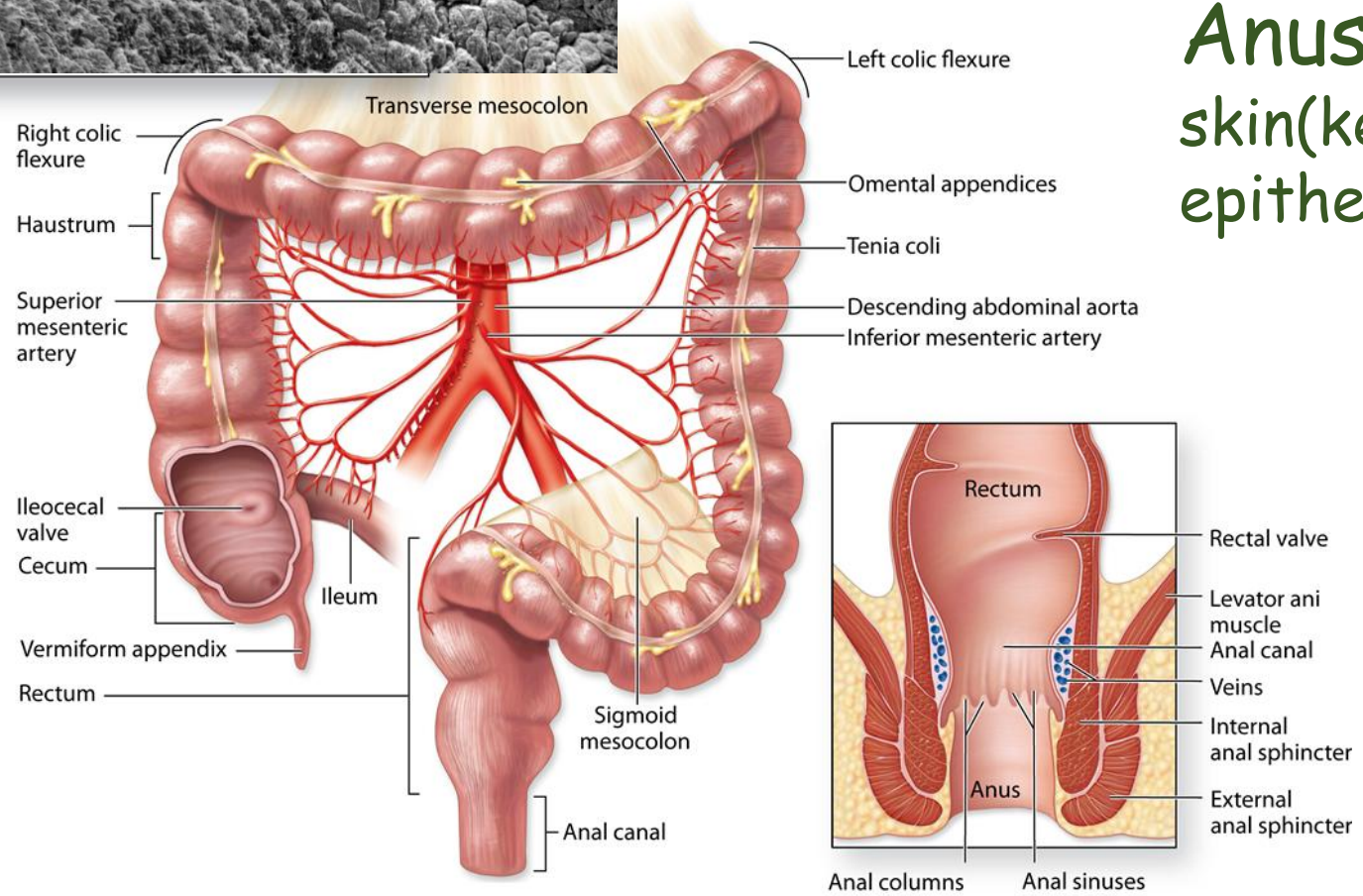
II.LARGE INTESTINE

SEM of the ileocecal junction



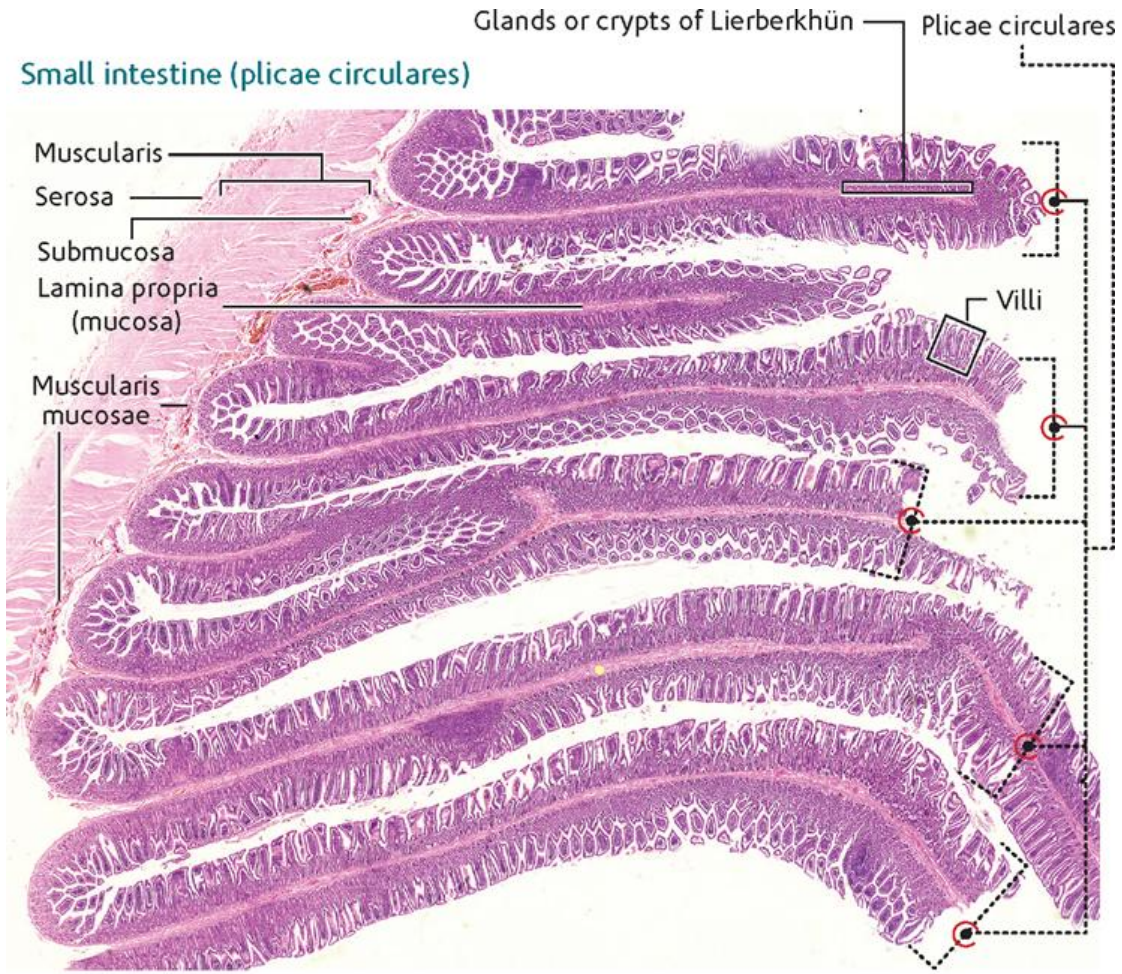
Begins at ileocecal junction, Appendix, Caecum, Colon: ascending, transverse, descending& sigmoid, Rectum: **upper anal canal & lower anal canal)** .

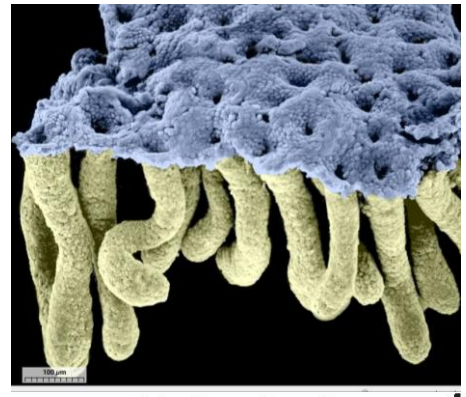
Anus: is covered by perianal skin(keratinized stratified squamous epithelium).

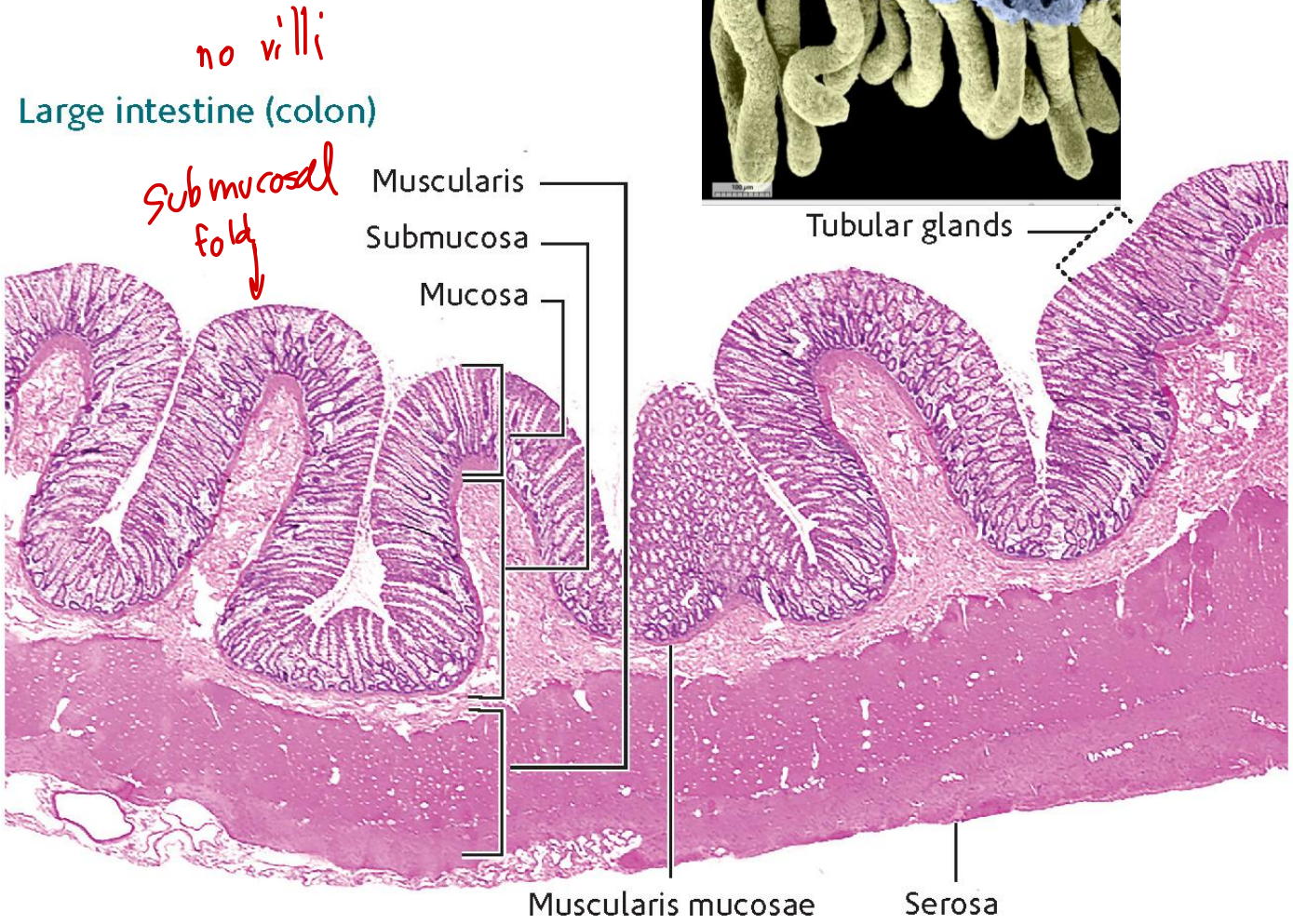


The primary function of the mucosa of the large intestine is the reabsorption of water, sodium, vitamins and minerals and elimination of un-digestion food.

COMPARISON BETWEEN SMALL AND LARGE INTESTINE

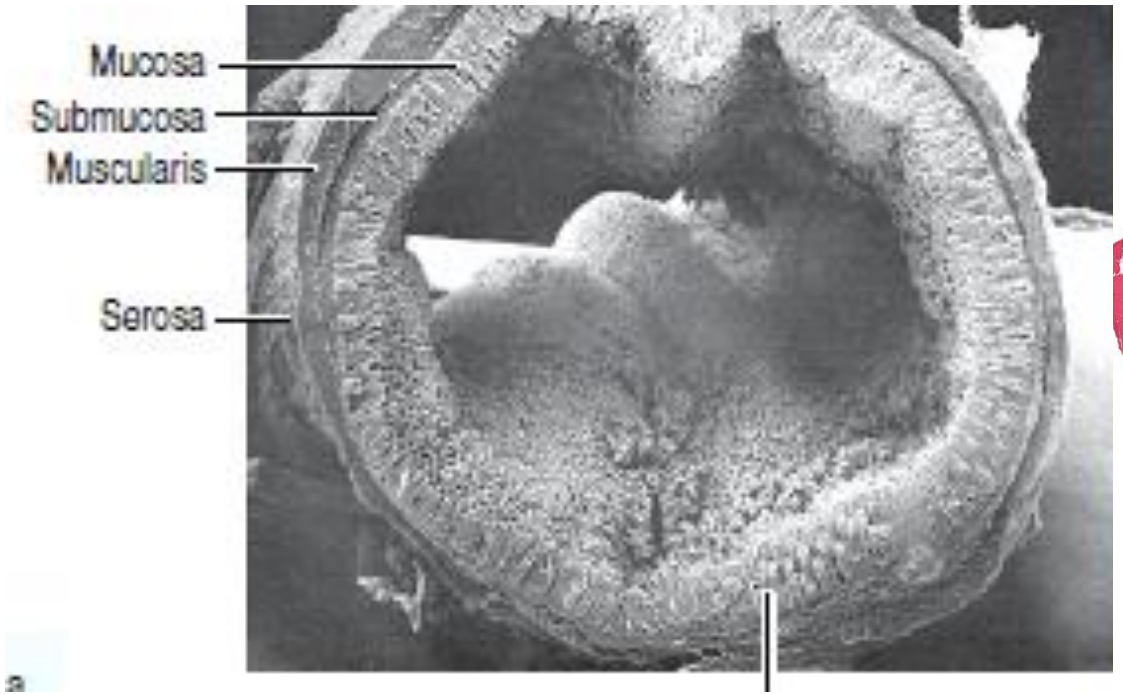




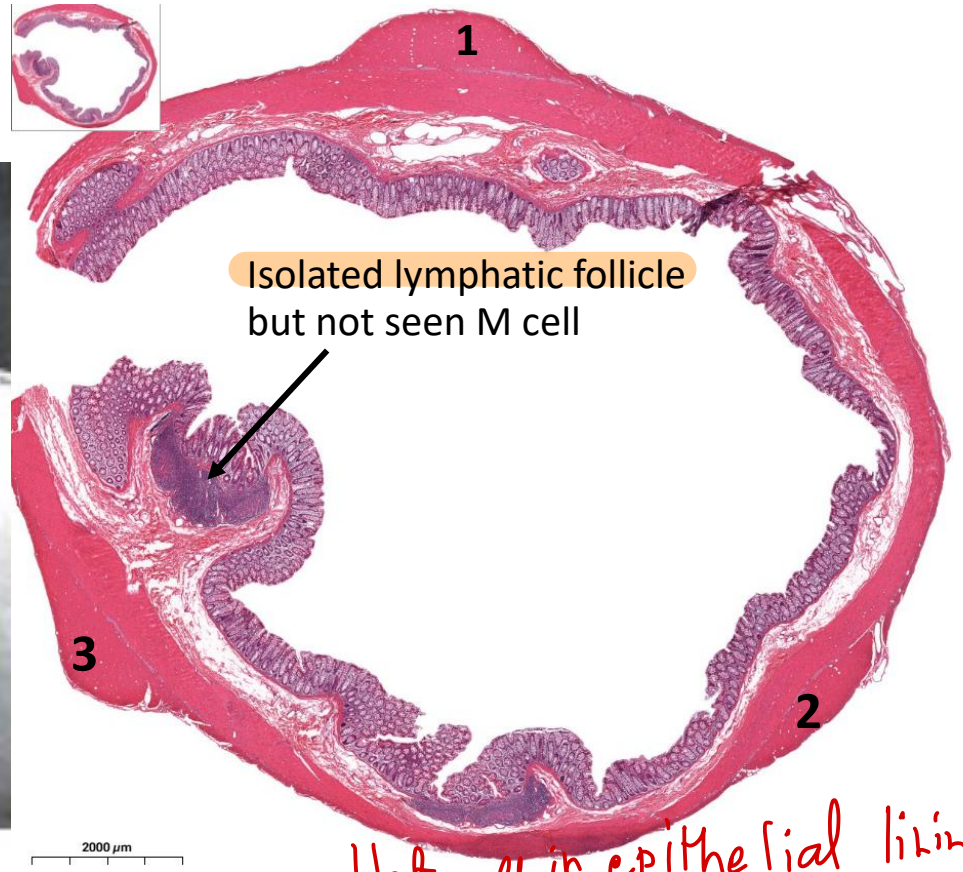


The muscularis is responsible for segmentation and peristaltic movement of the contents of the small intestine

SEM of LARGE INTESTINE



Tubular(intestinal) glands or crypts of Lieberkuhn

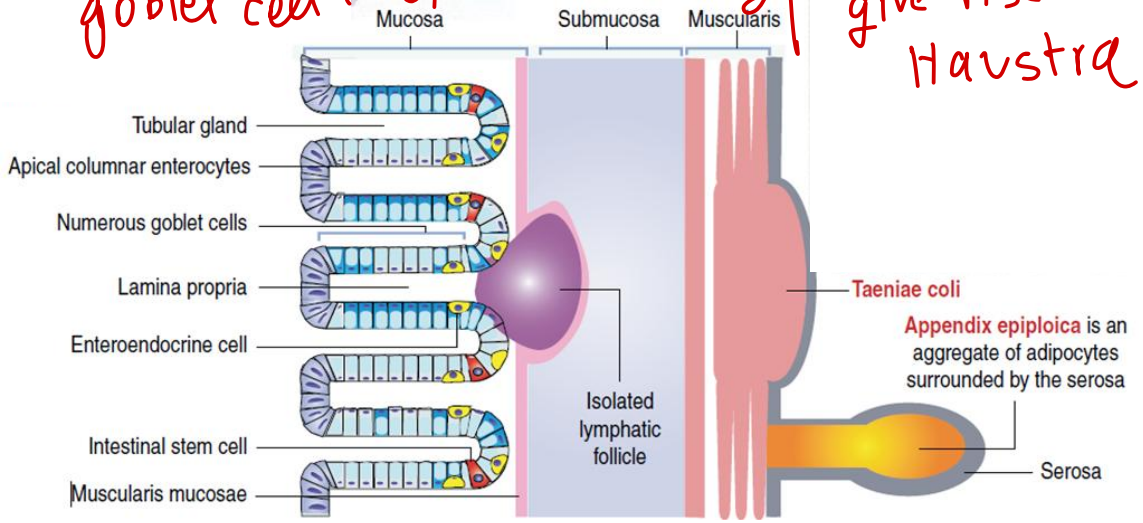


Fascicles of the *outer longitudinal layer aggregate into 3 spaced bands called teniae coli*. The contraction of the taenae coli & circular layer (thin) draws the colon into haustra.

goblet cell in epithelial lining

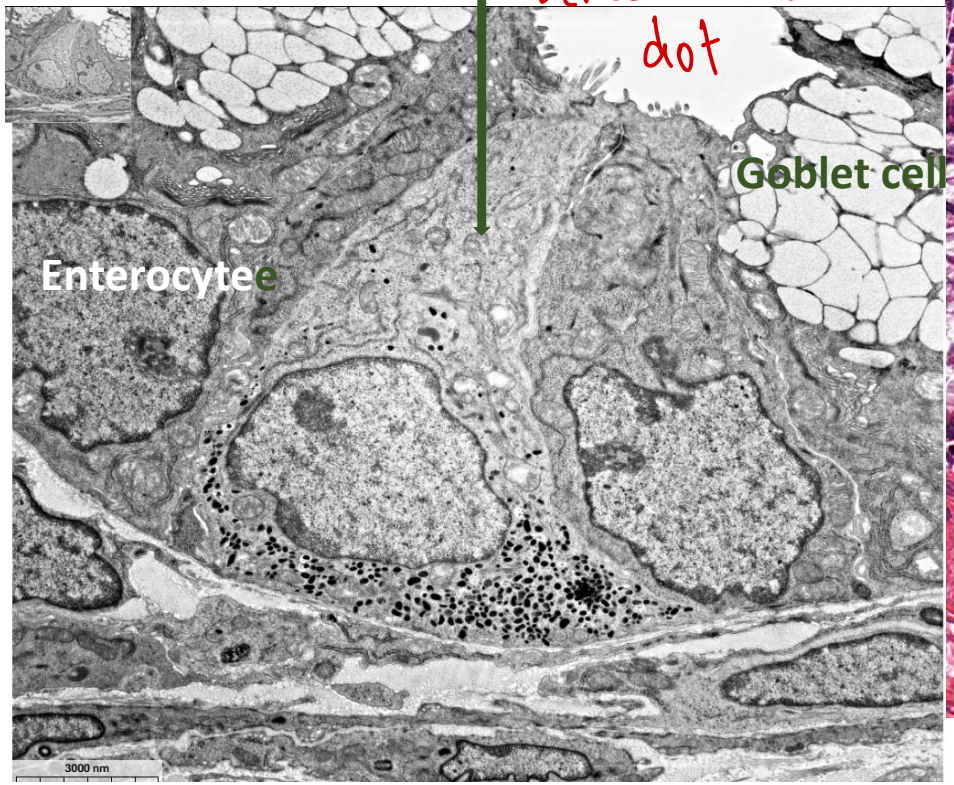
The presence of the teniae coli give rise to the Haustra

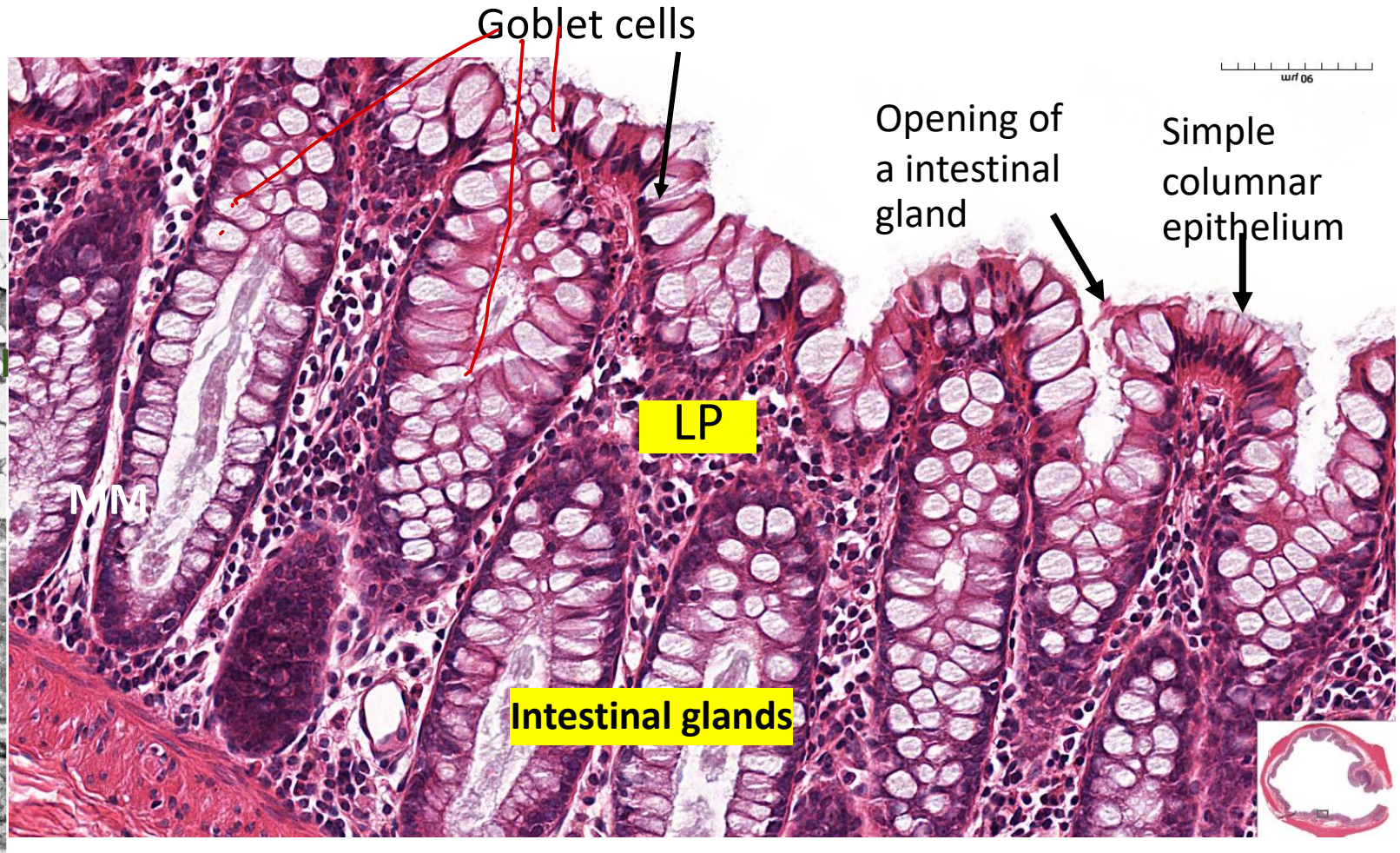
Straight tubular glands with abundant goblet cells & plicae circulares not beyond the ileocecal valve, no villi, taenae coli, mostly lymphatic nodules.



Mucosa of the large intestine is free of fold & with apical microvilli (short & a few)

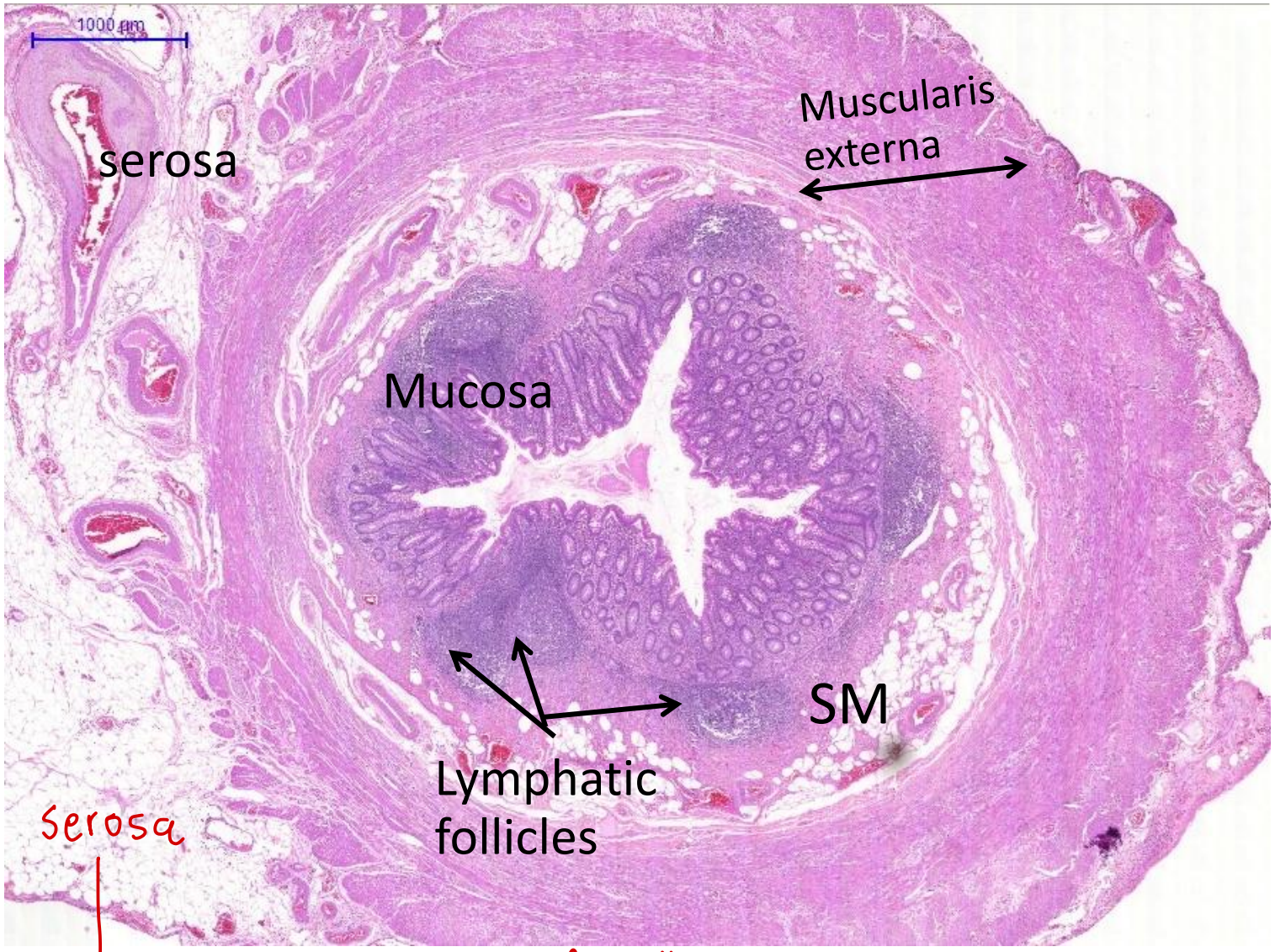
1) Simple columnar absorptive cells with short apical microvilli
2) Predominant goblet cells
3) Stem cells at the base of the tubular glands
4) Entero-endocrine cells





VERMIFORM APPENDIX is a diverticulum of the cecum and has layers similar to those of the large intestine.

- *MUCOSA :*
  - EPITHELIUM ; enterocytes with numerous goblet cells,
  - LAMINA PROPRIA: rich in lymphoid follicles and crypts of Lieberkuhn as Straight intestinal glands
  - Muscularis mucosae
- *SUBMUCOSA*
- *MUSCULARIS EXTERNA*
- *SEROSA :.as the visceral peritoneum*
  - consists of connective tissue covered by simple squamous epithelium, or mesothelium



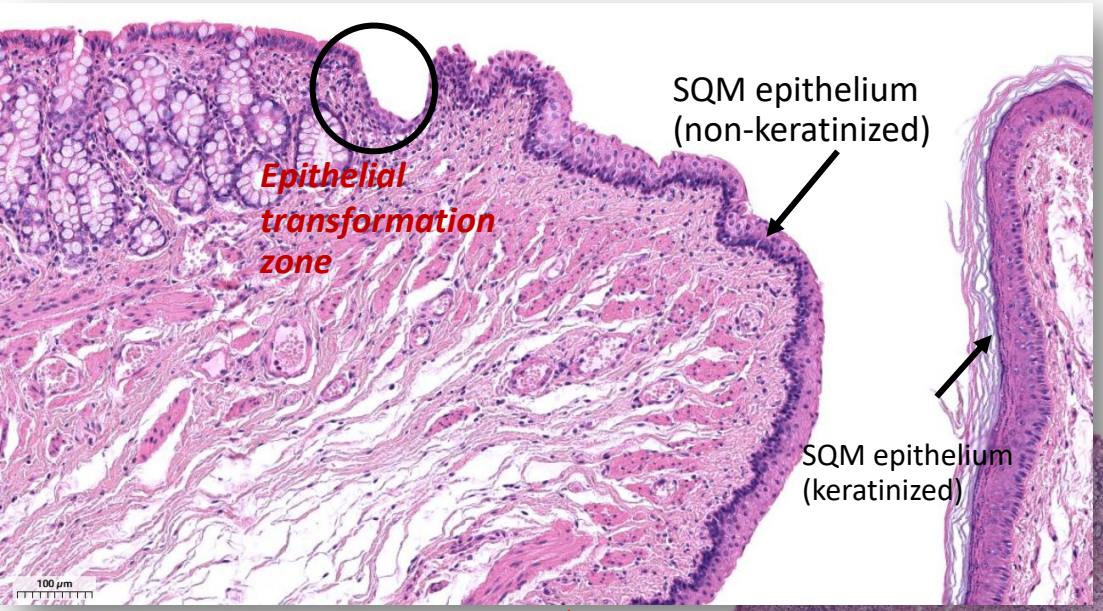
Recto-Anal Junction

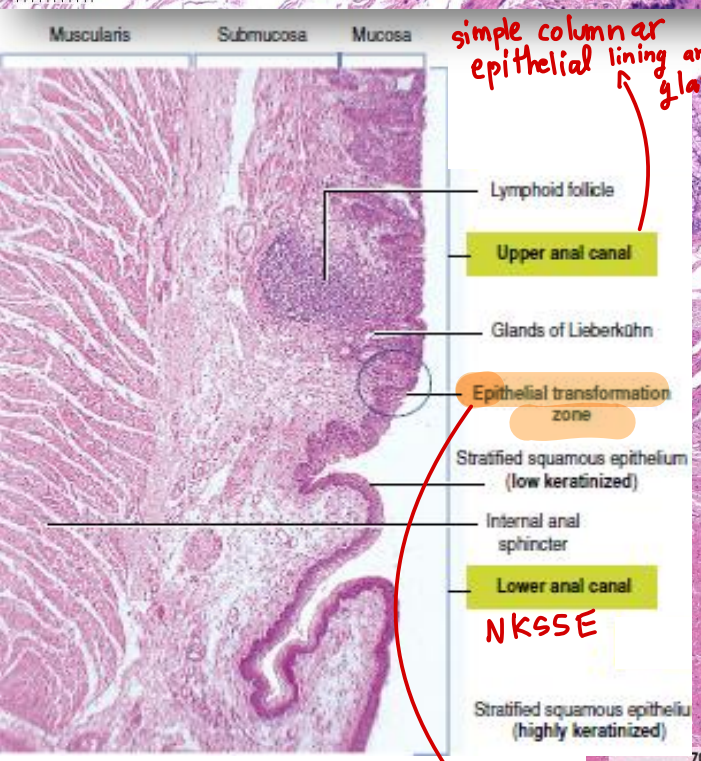
transformation zone (columnar → stratified)

- Rectum consists of (1) upper anal canal, (2) lower anal canal
- Anus

epithelial lining = keratinized stratified squamous epithelium (keratin present)

nonkeratinized stratified epithelium of the mucous type





simple columnar epithelial lining and glands

Sub-Mucosa

Intestinal gland

Internal sphincter

Anus

Submucosa contains *circumanal glands (i.e., apocrine sweat glands and sebaceous glands*

External sphincter part of the pelvic floor

NKSSE

columnar → stratified squamous epithelium

An abnormal form of constipation is seen in

Hirschsprung's disease (congenital megacolon)

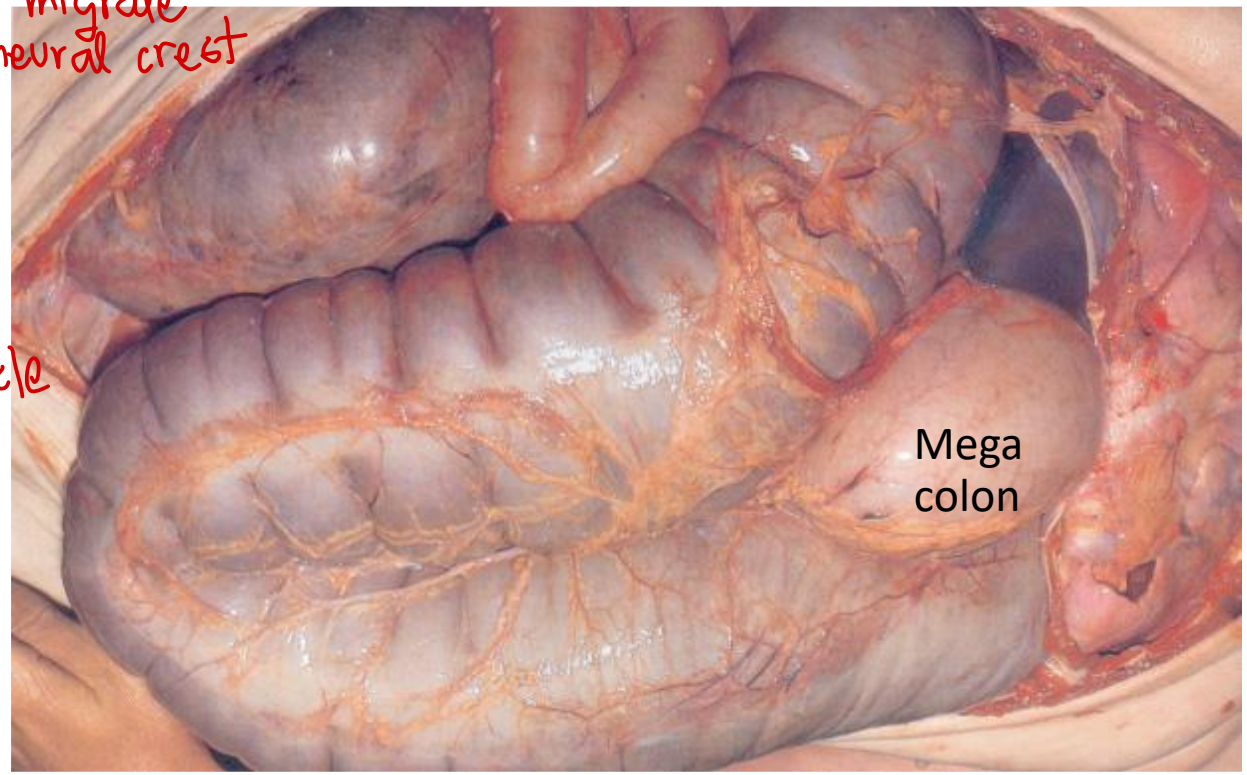
This condition, called **aganglionosis**, results from an **arrest in the migration of cells from the neural crest**, the precursors of the intramural ganglion cells of the plexuses of Meissner and Auerbach.

Absence of the enteric NS in the segment of the distal colon due to defects of neural cell migration and development. It is **caused by the mutations of the RET gene** encoding a receptor tyrosine kinase. RET signaling is required for migration of neural crest cells into the distal portions of the large intestine and their differentiation into intramural ganglion cells of the plexuses of Meissner and Auerbach of the enteric nervous system.

This condition is apparent shortly after birth when the abdomen of the infant becomes distended and little meconium is eliminated.

bc fail to migrate from neural crest

absent ganglion cells that's normally found in muscle layer



The diagnosis is confirmed by a biopsy of the mucosa and submucosa of the rectum, showing thick and irregular nerve bundles, abundant acetylcholinesterase detected by immunohistochemistry and a lack of ganglion cells.

- cause no movement of content within large intestine

Some CLINICAL CORRELATION OF THE LOWER GI TRACT

❖ **Malabsorption syndromes** are characterized by a deficit in the absorption of fats, proteins, carbohydrates, salts and water by the mucosa of the small intestine. Malabsorption syndromes can be caused by:

1.*Abnormal digestion of fats and proteins* by pancreatic diseases (pancreatitis or cystic fibrosis) or lack of solubilization of fats by defective bile secretion (hepatic disease or obstruction of bile flow into the duodenum).

2.*Enzymatic abnormalities at the brush border* , where disaccharidases and peptidases cannot hydrolyze carbohydrates (lactose intolerance) and proteins.

3.*A defect in the transepithelial transport by enterocytes* .

❖ Inflammatory bowel disease : Ulcerative colitis(can affect the mucosa of the large intestine) whereas Crohn's disease affects any segment of the intestinal tract. Both are clinically characterized by diarrhea, pain, and periodic relapses.

❖ Hemorrhoids : veins connecting the internal and external hemorrhoidal plexuses can become dilated and bulge the mucosa.

❖ Colorectal tumors (determined by a defect in APC gene) develop from a polyp. Some polyps are non-neoplastic and re relatively common in person 60 years and older. Familial polyposis syndrome and the Peutz-Jeghers syndrome possess polyps can be present in large number.

❖ Colorectal cancer occurs at a younger age (45 to 60 years of age) among patients who have the **Lynch syndrome** than among patients who have sporadic colorectal cancer (69 years of age).

Answers for some clinical questions for lower G (7 items)

1. This group of orders is characterized by decreased absorption, most often in the small bowel, of essential nutrients. Patients can present with chronic diarrhea and/or steatorrhea(fatty stool) as well as with systemic effects, such as weight loss and specific vitamin and mineral deficiencies, particularly the lipid-soluble vitamins A, D, E, and K.
2. Diarrhea
3. Malabsorption syndromes can be caused by:

1)Abnormal digestion of fats and proteins by pancreatic diseases (pancreatitis or cystic fibrosis) or lack of solubilization of fats by defective bile secretion (hepatic disease or obstruction of bile flow into the duodenum).

2)Enzymatic abnormalities at the brush border , where disaccharidases and peptidases cannot hydrolyze carbohydrates (lactose intolerance) and proteins.

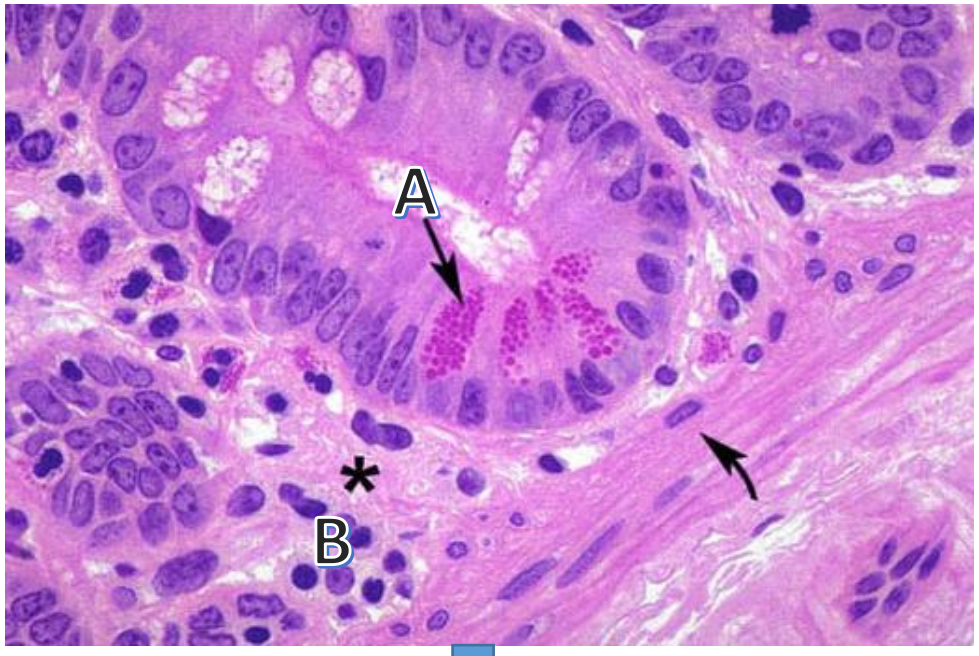
3)A defect in the transepithelial transport by enterocytes .

4. Ulcerative colitis and Crohn's disease, both are clinically characterized by diarrhea, pain, and periodic relapses.
5. It is Gluten-Sensitive Enteropathy, i. e., autoimmune entity with antibodies against the water-insoluble gliadin fraction of gluten, a protein found in wheat, barley, rye, and oats.
6. It is the congenital megacolon which is characterized by complete function obstruction of the large bowel due to the absence of ganglionic cells of both the submucosal and myenteric neural plexuses.

The diagnosis is confirm by a biopsy of the mucosa and submucosa of the rectum showing thick and irregular nerve bundles, *abundant acetylcholinesterase detected by immunohistochemistry and a lack of ganglion cells.*

7. A Polyp, but some polyps are not non-neoplastic

Laboratory Quiz for the lower GI(3 items)



1 A. What is the significance of these granules A?

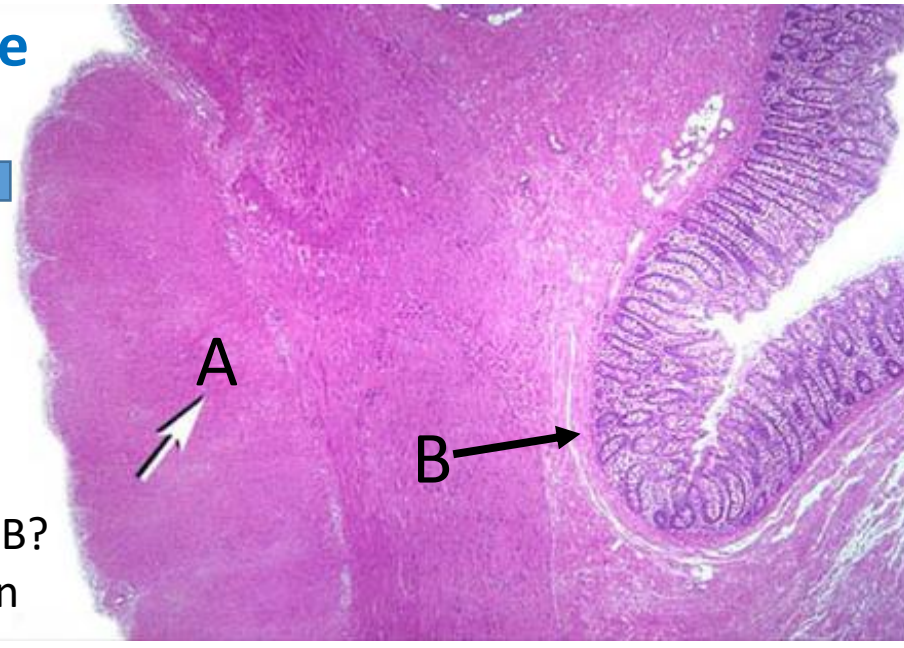
a) Involve in the cellular immune surveillance of the intestinal tract
b) Involve in Megaloblastic anemia
c) Involve in intestinal mucosa blanket
d) Involve in the gastric mucosa blanket

B. Which is **in**correct for area B?

a) It is known as lamina propria
b) It also occupies in the cores of microvilli
c) It surrounds crypts of intestinal glands
d) It is the loose areolar connective tissue

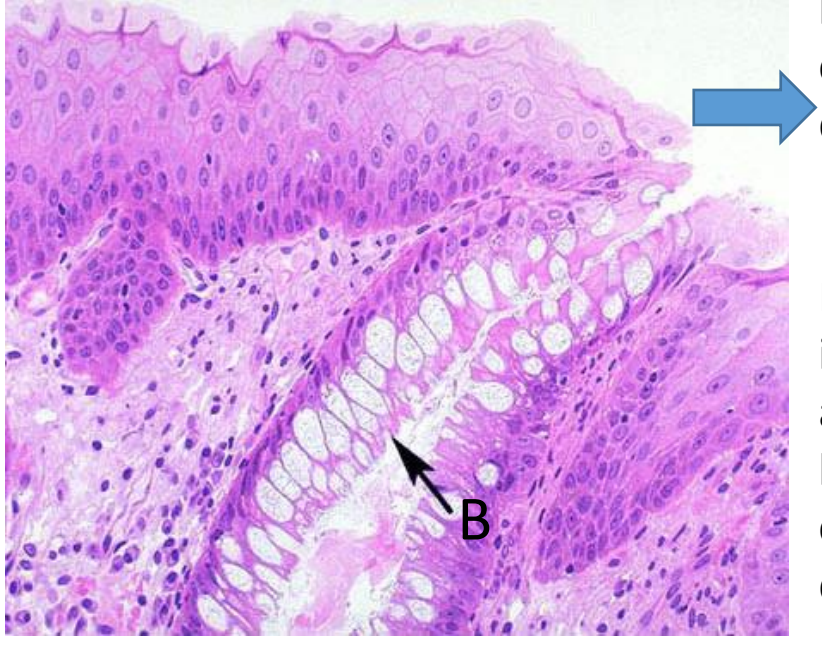
2.A.Identify structure A

a)Inner circular muscularis
b) Taenae coli
c)Covering with adventitia
d)Muscularis of small intestine



B. Which is correct for structure B?

a)It contracts reducing the lumen
b)It belongs to submucosa
c) It functions for mucosal motility
d) It is the lamina propria



3. A. Identify the whole picture

a) Esophago-cardiac junction
b) Anorectal junction
c) Gastroduodenal junction
d) Pharyngo-esophageal junction

B. Which cell type is **not** located in the epithelium B?

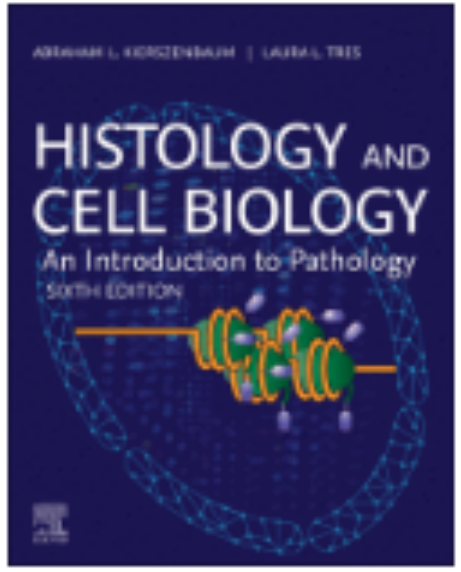
a) Goblet cells
b) Enteroendocrine cells
c) Stem cells
d) Paneth cells

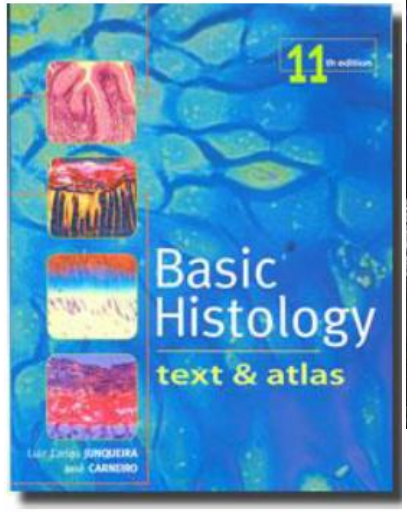
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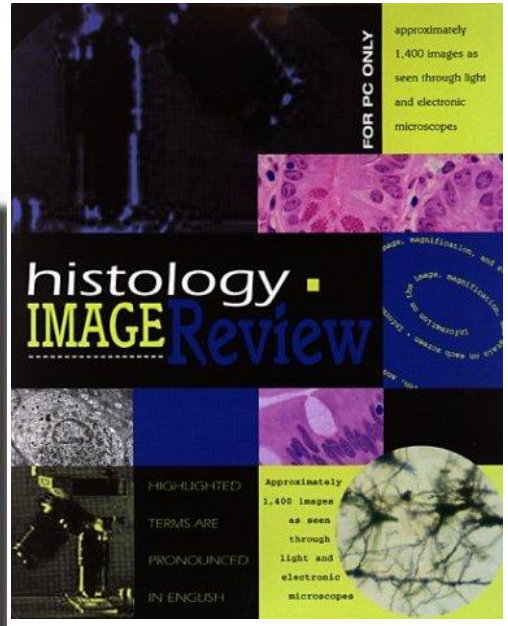
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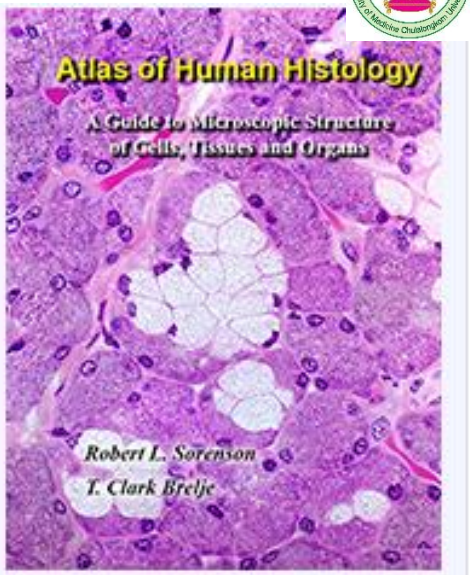


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